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Cochrane Database of Systematic Reviews 2026, Issue 7. Art. No.: CD015898.
DOI: 10.1002/14651858.CD015898.pub2.

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[Intervention Review]

Larger versus smaller red blood cell volume per transfusion in hospitalized adults, children and preterm neonates

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Editorial group: Cochrane Central Editorial Service.
Publication status and date: New, published in Issue 7, 2026.

Citation: Roumeliotis N, Sabbagh G, Belkacem M, Dodin P, Du Pont-Thibodeau G, Callum J, Tucci M, Carrier FM, Lacroix J. Larger versus smaller red blood cell volume per transfusion in hospitalized adults, children and preterm neonates. *Cochrane Database of Systematic Reviews* 2026, Issue 7. Art. No.: CD015898. DOI: 10.1002/14651858.CD015898.pub2.

A B S T R A C T

Rationale

While substantial research has evaluated the safety of hemoglobin thresholds and storage length for red blood cell (RBC) transfusion, there is minimal literature on RBC transfusion volume per transfusion event in hospitalized patients. The Choosing Wisely initiative has made a recommendation to transfuse a single unit of RBC per transfusion event, although this is based on little evidence. No recommendation has been made for pediatrics. This review evaluates the effect of the volume of RBC transfusion, administered when the decision has been made to transfuse a hospitalized patient, on mortality and other important outcomes.

Objectives

To compare the effectiveness and safety of larger versus smaller RBC volume per transfusion for anemia in hospitalized adults, children, and preterm neonates.

Search methods

We searched Evidence-Based Medicine Reviews (EBMR; including CENTRAL), MEDLINE, Embase, Web of Science, and other databases on 5 August 2025, with reference checking, citation searching and contacting study authors to identify additional studies.

Eligibility criteria

Eligible studies were randomized controlled trials (RCTs) and non-randomized studies of interventions (NRSIs) that included adults, children or neonates and compared a larger volume of RBC per transfusion event (intervention) to a smaller volume of RBC transfusion (control). We defined a transfusion event as a single administration of blood products within six hours.

Outcomes

The critical outcome was mortality. Important outcomes were length of hospital stay, hospital-free days, transfusion-associated adverse events (TAAE), organ dysfunction, number of RBCs transfused, rebleeding and allogeneic donor exposure.

Risk of bias

We assessed risk of bias using the Cochrane RoB 2 tool for RCTs and the Risk of Bias in Non-Randomized Studies of Interventions (ROBINS-I) version 2 tool for NRSIs.

Synthesis methods

Two authors independently extracted data from included studies and assessed the risks of bias. We analyzed data from the included RCTs and NRSIs separately. We used the risk ratio (RR) and mean difference (MD) for pooled effects, using a random-effects model to account for heterogeneity between studies. We used GRADE to assess the certainty of evidence.

Included studies

We included 12 studies (5478 participants); five RCTs, and seven controlled NRSIs. Nine studies included adults (1945 participants), one included children (3199 participants) and two included preterm neonates (334 participants). Eight of the adult studies' participants were in hematology wards or bone marrow transplant units, and one study included postpartum participants. The pediatric study included anemic children, and the neonatal studies included preterm infants < 32 weeks' gestation or weighing < 1.5 kg. All adult studies compared two units of RBCs to one unit of RBCs per transfusion event. The pediatric study compared 15 mL/kg to 10 mL/kg RBCs, and the neonatal studies compared 20 mL/kg to 10 mL/kg RBCs. The included RCTs ranged from low to high risk of bias and the NRSIs ranged from low to critical risk of bias.

Synthesis of results

Nine studies reported on mortality, although the definition of time of death varied. Meta-analysis of adult studies demonstrated no difference in mortality between the two-unit RBC transfusion group and the one-unit RBC transfusion group in RCTs (RR 1.29, 95% CI 0.62 to 2.67; 2 RCTs, 322 participants; low-certainty evidence) or in NRSIs (RR 1.03, 95% CI 0.62 to 1.71; 5 NRSIs, 1198 participants; low-certainty evidence).

There was no difference in hospital length of stay between the two-unit RBC group and the one-unit RBC group in RCTs (MD 0.08, 95% CI −0.66 to 0.82; 2 RCTs, 311 participants; low-certainty evidence) and in NRSIs (MD −0.15, 95% CI −1.68 to 1.38; 4 NRSIs, 1048 participants; very low-certainty evidence).

No studies reported hospital-free days as an outcome.

There was no difference in TAAEs between the two-unit RBC group and the one-unit RBC group for RCTs (RR 1.25, 95% CI 0.61 to 2.56; 3 RCTs, 388 participants; low-certainty evidence) or NRSIs (RR 0.74, 95% CI 0.30 to 1.82; 3 NRSIs, 546 participants; very low-certainty evidence).

During their hospital stay, participants in the two-unit RBC transfusion group received more RBC units than those in the one-unit RBC transfusion group in the NRSIs (MD 0.65 units, 95% CI 0.55 to 0.75; 4 NRSIs, 1064 participants; low-certainty evidence). However, no difference was seen between groups in the RCTs (MD 0.90 units, 95% CI 0.68 to 1.11; 3 RCTs, 378 participants; low-certainty evidence).

During each transfusion event, patients in the two-unit RBC transfusion group received 0.66 RBC units more compared to those in the one-unit RBC group (MD 0.66 units, 95% CI 0.59 to 0.73; 3 NRSIs, 791 participants), reflecting adherence to the intervention.

There was no difference between the two RBC transfusion groups' rates of thrombosis reported in RCTs (RR 2.26, 95% CI 0.51 to 9.95; 2 RCTs, 311 participants; very low-certainty evidence). No NRSIs reported rates of thrombosis. There was no difference between the two RBC transfusion groups' rebleeding rates in either RCTs (RR 0.52, 95% CI 0.16 to 1.68; 1 RCT, 245 participants; moderate-certainty evidence) or NRSIs (RR 0.64, 95% CI 0.36 to 1.14; 3 studies, 585 participants; very low-certainty evidence).

No studies reported on allogenic donor exposure.

Authors' conclusions

In adults, when comparing two units of RBCs (larger volume) to one unit of RBCs (smaller volume) for a single transfusion event, there was no difference in mortality, length of hospital stay or TAAEs, albeit with low or very low-certainty evidence. However, it reduced the number of required RBC units.

The results indicate that a larger transfusion volume confers no clinical benefit over a smaller, more restrictive transfusion strategy (lower transfusion volume), but may lead to a higher amount of blood administered. Given the adverse events associated with RBC transfusion and the resource limitation of the allogeneic blood supply, it is reasonable to support the recommendations for single unit RBC transfusion per transfusion event in adults.

In children, evidence is still too limited to be able to make a recommendation on an RBC volume.

Funding

This study did not receive funding.

Registration

This review protocol was previously published with the Cochrane database of systematic reviews (DOI 10.1002/14651858.CD015898).

PLAIN LANGUAGE SUMMARY

What are the benefits and harms of giving a larger or smaller amount of red blood cells during a blood transfusion?

Key messages

- Giving adults smaller amounts of red blood cells during a blood transfusion may not make any difference to their risk of dying, how long they stay in hospital, or any unwanted treatment effects, compared with giving larger amounts of red blood cells, but the evidence from some studies is unclear.
- At the same time, it may reduce the amount of blood given for one transfusion, which saves money and means that more blood is available for other people who need a transfusion.
- We did not find enough information to know whether it is better to give children or babies who need a transfusion larger or smaller amounts of red blood cells.

What is a blood transfusion?

Many people in hospital do not have enough healthy red blood cells to carry oxygen around their body, which can make them tired, weak, and short of breath. This is called anemia, and it can be treated in several different ways. Sometimes, people may need to have extra blood added to their own (a blood transfusion). Red blood cells are an important part of the extra blood. However, there is not much information available for doctors about the amount of red blood cells (the 'red blood cell volume') that should be given.

The volume of red blood cells in the blood can be measured in units (e.g. one unit of red blood cells) for adults. For babies and children, doctors give a certain amount depending on how much the child weighs. For example, they might give 20 mL of red blood cells for every kg the child weighs (20 mL/kg). Because blood transfusion involves some risks and there is a limited amount of blood available, doctors have been recommended to give the smallest effective amount to adults. Currently, it is recommended that one unit of red blood cells (rather than two units) is given to adults in hospital who are not bleeding and whose condition is not changing quickly. In children and newborns, there is no recommended standard amount for transfusion.

What did we want to find out?

We wanted to find out if it is safe to give a smaller amount of red blood cells to adults, children and newborns in hospital who need a blood transfusion. We looked at whether people stayed in hospital longer, had any unwanted effects from the treatment, any problems with their heart, breathing or other body systems, if they started bleeding again, and how many red blood cells were given.

What did we do?

We searched for studies that compared two different volumes of red blood cell transfusion for people in hospital. For example, two units of red blood cells compared with one unit of red blood cell for adults, or 20 mL/kg compared with 10 mL/kg for children. We compared and summarized the results of the studies and rated our confidence in the evidence, based on factors such as study methods and sizes.

What did we find?

We found 12 studies that involved 5478 people. Nine studies included a total of 1945 adults; one included women needing a blood transfusion after childbirth and the others included people who needed a blood transfusion because of blood cancer. One study included 3199 anemic children aged two months to 12 years, and two studies included a total of 334 newborn babies born before 32 weeks of pregnancy.

In adults, when comparing one unit of red blood cells to two units of red blood cells for a single blood transfusion, there may be no increased risk of dying, no increase in length of hospital stay, and no increase in any unwanted effects of the transfusion, blood clots (thrombosis) or starting to bleed again. However, the evidence is unclear.

What are the limitations of the evidence?

We have little confidence in most of the evidence we found because there were not enough studies to be certain about the results. In some of the studies, it is possible that people were aware of which treatment patients were getting. People in some of the studies were not randomly placed into the different treatment groups. This means that differences between the groups could be due to differences between people rather than between the treatments.

How up to date is this evidence?

The evidence is current to August 2025.

SUMMARY OF FINDINGS

Summary of findings 1. Summary of findings table - Two units of red blood cell volume compared to one unit of red blood cell volume for anemia in hospitalized adults (randomized controlled trials)

Two units of red blood cell volume compared to one unit of red blood cell volume for anemia in hospitalized adults (randomized controlled trials)

Patient or population: anemia in hospitalized adults (randomized controlled trials)
Setting: hospital
Intervention: two units of red blood cell volume
Comparison: one unit of red blood cell volume

Outcomes	Anticipated absolute effects* (95% CI)		Relative effect (95% CI)	№ of participants (studies)	Certainty of the evidence (GRADE)	Comments
	Risk with one unit of red blood cell volume	Risk with two units of red blood cell volume				
Mortality[a]	74 per 1000	**95 per 1000** (46 to 197)	**RR 1.29** (0.62 to 2.67)	322 (2 RCTs)	⊕⊕⊝⊝ Low[b,c]	
Length of hospital stay	The mean length of hospital stay was **5.03** days	MD **0.08 days higher** (0.66 lower to 0.82 higher)	-	311 (2 RCTs)	⊕⊕⊝⊝ Low[c,d]	
Hospital-free days	Not pooled	Not pooled	Not pooled	(0 RCTs)	-	Not pooled
Transfusion-associated adverse event	61 per 1000	**77 per 1000** (37 to 157)	**RR 1.25** (0.61 to 2.56)	388 (3 RCTs)	⊕⊕⊝⊝ Low[b,e]	
Thrombotic event	13 per 1000	**29 per 1000** (6 to 126)	**RR 2.26** (0.51 to 9.95)	311 (2 RCTs)	⊕⊝⊝⊝ Very low[e,f]	
RBC units transfused per patient	The mean RBC units transfused per patient was **1.94** units	MD **0.9 units higher** (0.68 higher to 1.11 higher)	-	378 (3 RCTs)	⊕⊕⊝⊝ Low[b,e]	
Rebleeding	64 per 1000	**33 per 1000** (10 to 108)	**RR 0.52** (0.16 to 1.68)	245 (1 RCT)	⊕⊕⊕⊝ Moderate[c]	

***The risk in the intervention group** (and its 95% confidence interval) is based on the assumed risk in the comparison group and the **relative effect** of the intervention (and its 95% CI).

CI: confidence interval; **MD:** mean difference; **RR:** risk ratio

GRADE Working Group grades of evidence
High certainty: we are very confident that the true effect lies close to that of the estimate of the effect.
Moderate certainty: we are moderately confident in the effect estimate: the true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different.
Low certainty: our confidence in the effect estimate is limited: the true effect may be substantially different from the estimate of the effect.
Very low certainty: we have very little confidence in the effect estimate: the true effect is likely to be substantially different from the estimate of effect.

See interactive version of this table: https://gdt.gradepro.org/presentations/#/isof/isof_question_revman_web_465686487024853629.

[a] RCT: randomized controlled trial
[b] Downgraded one level as risk of bias for one or more RCTs is high.
[c] Downgraded one level for imprecision for wide and non-precise 95% CIs in one or more studies.
[d] Downgraded one level for imputation of data from median to mean for one or more studies.
[e] Downgraded one level for indirectness given different populations.
[f] Downgraded two levels for imprecision due to very wide confidence intervals in more than one study.

Summary of findings 2. Summary of findings table - Two units of red blood cell volume compared to one unit of red blood cell volume for anemia in hospitalized adults (non-randomized studies)

Two units of red blood cell volume compared to one unit of red blood cell volume for anemia in hospitalized adults (non-randomized studies)

Patient or population: anemia in hospitalized adults (non-randomized studies)
Setting: hospital
Intervention: two units of red blood cell volume
Comparison: one unit of red blood cell volume

Outcomes	Anticipated absolute effects* (95% CI)		Relative effect (95% CI)	№ of participants (studies)	Certainty of the evidence (GRADE)	Comments
	Risk with one unit of red blood cell volume	Risk with two units of red blood cell volume				
Mortality	46 per 1000	**47 per 1000** (28 to 78)	**RR 1.03** (0.62 to 1.71)	1198 (5 non-randomised studies)	⊕⊕⊝⊝ Low[a,b]	
Length of hospital stay	The mean length of hospital stay was **19.97** days	MD **0.15 days lower** (1.68 lower to 1.38 higher)	-	1048 (4 non-randomised studies)	⊕⊝⊝⊝ Very low[a,b,c]	
Hospital-free days	The mean hospital-free days was **0**	Not pooled	-	(0 non-randomised studies)	-	Not pooled

Transfusion-associated adverse event	39 per 1000	**29 per 1000** (12 to 71)	**RR 0.74** (0.30 to 1.82)	546 (3 non-randomised studies)	⊕⊝⊝⊝ Very low[a,d]	
Thrombotic event	Not pooled	Not pooled	Not pooled	(0 studies)	-	Not pooled
RBC units transfused per patient	The mean RBC units transfused per patient was **1.49** units	MD **0.65 units higher** (0.55 higher to 0.75 higher)	-	1064 (4 non-randomised studies)	⊕⊕⊝⊝ Low[a,b]	
Rebleeding	105 per 1000	**67 per 1000** (38 to 120)	**RR 0.64** (0.36 to 1.14)	585 (3 non-randomised studies)	⊕⊝⊝⊝ Very low[b,e]	

***The risk in the intervention group** (and its 95% confidence interval) is based on the assumed risk in the comparison group and the **relative effect** of the intervention (and its 95% CI).

CI: confidence interval; **MD:** mean difference; **RR:** risk ratio

GRADE Working Group grades of evidence
High certainty: we are very confident that the true effect lies close to that of the estimate of the effect.
Moderate certainty: we are moderately confident in the effect estimate: the true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different.
Low certainty: our confidence in the effect estimate is limited: the true effect may be substantially different from the estimate of the effect.
Very low certainty: we have very little confidence in the effect estimate: the true effect is likely to be substantially different from the estimate of effect.

See interactive version of this table: https://gdt.gradepro.org/presentations/#/isof/isof_question_revman_web_465773845712160562.

[a] Downgraded one level as risk of bias for non-randomized studies is serious.
[b] Downgraded one level for imprecision given wide 95% confidence intervals in one or more studies.
[c] Downgraded one level for indirectness due to imputation of data from median to mean for one or more studies.
[d] Downgraded two levels for imprecision due to very wide confidence intervals in more than one study.
[e] Downgraded two levels as risk of bias for non-randomized studies is very serious.

BACKGROUND

Description of the condition

Anemia in hospitalized patients is common and may be associated with multiple long-term morbidities. Anemia may be caused by congenital diseases (e.g. sickle cell disease, thalassemia) or acquired diseases (e.g. blood loss, cancer, hemolysis, iron deficiency). In many hospitalized patients, anemia is caused by blood loss, diagnostic phlebotomy, or inflammation-related anemia. Hospital management of anemia ranges from strategies to reduce phlebotomy, reduction of surgical and non-surgical bleeding, and increasing production of red blood cells (RBCs), with RBC transfusion as a last resort. This approach is referred to as Patient Blood Management (PBM) by the World Health Organization (WHO), the pillars of which are detecting and managing anemia, minimizing blood loss, optimizing the tolerance to anemia and minimization of RBC transfusions [1]. RBC transfusion has its own inherent risks, including transfusion reactions, volume overload, pulmonary complications and organ dysfunction [2, 3], therefore the risks must be weighed against the benefits. Furthermore, RBCs are a scarce and costly resource that must be managed sparingly [4, 5, 6, 7, 7, 8]. Given these risks and limited resources, and no proven benefits of liberal transfusion practices, minimizing RBC transfusions has become an important pillar of PBM for hospitalized patients. Minimizing RBC transfusions is achievable through different approaches, such as using low restrictive hemoglobin thresholds and limiting the volume or number of transfusions when a transfusion decision has been made. This shift toward more restrictive transfusion approaches has occurred over the last two decades in clinical practice.

Description of the intervention and how it might work

Allogeneic blood transfusion transiently corrects anemia by increasing the quantity of circulating RBCs, thus improving the body's oxygen-carrying capacity, but also increasing the risk of immediate transfusion reactions such as volume overload. Higher doses may be beneficial by more aggressively increasing oxygen-carrying capacity, but may also increase the risk of transfusion complications. There is no evidence-based consensus on the optimal number of RBCs needed to achieve clinical improvement in anemic patients, while reducing the associated risks of RBC transfusion [9, 10].

Pulmonary complications, such as transfusion-associated circulatory overload (TACO) and transfusion-related acute lung injury (TRALI), are among the most significant morbidities associated with RBC transfusion. TACO is characterized by respiratory deterioration due to volume overload and hydrostatic pulmonary edema, ensuing in the 12 hours after a transfusion, while TRALI is characterized by respiratory deterioration due to hyperpermeability pulmonary edema [11]. Hemovigilance reporting systems and a recently revised definition of TACO have increased the number of detected cases and awareness of the condition; however, it may be difficult to distinguish TACO from other respiratory complications after transfusion [11, 12]. A restrictive volume of transfusion favors a resource-sparing strategy that may reduce unnecessary risks of transfusion reactions in the patient, such as TACO, TRALI and other pulmonary complications, while reducing the utilization of costly transfusion.

There are currently no recommendations for a specific volume of RBC transfusion based on the etiology of anemia. In their transfusion medicine practice recommendations, the Choosing Wisely initiative recently made a recommendation for single-unit RBC transfusion in non-actively bleeding hospitalized anemic adults [9]. This recommendation is based on evidence of morbidity related to total amounts of blood received in studies of transfusion threshold and is supported by the Association for the Advancement of Blood and Biotherapies (AABB) [9, 13].

There is a large variation in the stated volumes of RBC transfusions used for anemic, non-bleeding hospitalized children [14]. The Australian PBM guidelines recommend an RBC transfusion volume based on weight and desired target hemoglobin [15]. While there is little evidence on what the target hemoglobin should be in non-bleeding infants and children, a 4 mL/kg RBC volume leads to an approximate 10 g/L hemoglobin rise, making 15 mL/kg of RBC suitable for most stable children weighing less than 20 kg to increase hemoglobin by at least 30 g/L [15, 16]. A single unit is recommended for most children weighing over 20 kg [15]. These recommendations are largely based on expert opinion given the substantial lack of evidence [16, 17, 18]. However, with the transition to single-unit transfusions in adults, a 290 mL RBC volume in a 70 kg patient is equivalent to a dose of 4 mL/kg, leading to a divergent practice compared to the commonly recommended 10 to 20 mL/kg in pediatrics [16]. This may be placing children at higher risk of avoidable transfusion reactions, particularly TACO.

In this review, the studied intervention was a larger volume of allogeneic RBC transfusion for a given transfusion event of an anemic patient compared to a smaller transfusion volume (for example, one versus two units in adults, and < 10 mL/kg and ≥ 10 mL/kg in children/neonates). We defined a transfusion event as a single transfusion within a six-hour period, or more than one transfusion in the case of immediate consecutive RBC transfusions (e.g. two consecutive RBC units). We did not include the volumes (or quantities) of RBC transfusions given multiple hours apart.

Why it is important to do this review

Multiple randomized trials have compared a restrictive to a liberal hemoglobin threshold for allogeneic RBC transfusion in a variety of hospitalized patients, including critically ill adults [3], critically ill children [19], and people with traumatic brain injury [20], gastrointestinal bleeding [21], coronary bypass surgery [22], and cancer [23, 24], among others. A recently updated Cochrane review of randomized controlled trials (RCTs) compared the effect of a restrictive versus a liberal RBC transfusion strategy on 30-day mortality [25, 26]. While many of these individual studies found that the liberal transfusion group received more RBC transfusions overall, the Cochrane review found no difference in mortality or other clinical outcomes (cardiac events, stroke, thromboembolism) between the liberal and the restrictive group [25]. Importantly, none of the studies specifically evaluated the volume of RBCs per given transfusion event, but most recommended a single-unit transfusion. Four large RCTs evaluated the impact of the length of storage of RBC units on mortality of transfused hospitalized patients [27, 28, 29, 30]. Data on the volume per transfusion were not collected in any of the trials, but a 'one RBC unit per transfusion' policy was adopted in all four RCTs.

While substantial research has evaluated the safety of hemoglobin thresholds and storage length for RBC transfusion, there is minimal literature on the impact of RBC transfusion volume per transfusion event in hospitalized patients. This review evaluates the effect on hospital mortality of the volume of RBC transfusion, administered when the decision has been made to transfuse an anemic hospitalized patient with any hemoglobin threshold.

Given significant transfusion practice variation in adults and children, and in order to reduce potential adverse events and morbidities associated with higher transfusion volumes (including TRALI and TACO) and spare limited donor resources, evidence is needed to guide clinicians in the practice of lower transfusion volume per transfusion event. The results of this systematic review will potentially support the 'Choosing Wisely' recommendation of a single unit per transfusion and provide guidance on the recommended volume of RBCs to transfuse in children and neonates.

OBJECTIVES

To compare the effectiveness and safety of larger versus smaller RBC volume per transfusion for anemia in hospitalized adults, children, and preterm neonates.

METHODS

The protocol of this review was previously registered and published with the Cochrane database of systematic reviews [31].

Differences between protocol and review

There are no major differences between the planned protocol and the review that was conducted.

Planned outcomes and subgroups were limited due to the number and types of studies included, and their respective outcomes and populations. As 28-day mortality was reported more frequently than hospital mortality, we selected it as the first mortality definition. While we planned on a minimum of three studies for a meta-analysis, we conducted two analyses with only two studies.

Criteria for considering studies for this review

Types of studies

We included published RCTs (individual and cluster-RCTs) and non-randomized studies of interventions (NRSI) in hospitalized adults, children, and neonates with any etiology or cause of anemia, published as full-text studies (not abstracts) in English or French.

NRSIs included before-after studies, interrupted time series, interventional cohorts, cross-sectional studies, case-control studies that compared two volumes of transfusion, and quasi-randomized studies. Quasi-randomized studies are experimental studies that test the causal effect of an intervention by comparing an intervention group with a control group. Unlike randomized studies, participants are assigned to groups using a non-random method. The inclusion of NRSIs in this review was based on the following.

1. Likely insufficient available evidence in randomized trials that address the study intervention.
2. Recommendations for the current use of one-unit RBC per transfusion ('Choosing Wisely') make undertaking studies randomizing patients to one versus two units of RBC ethically difficult (intervention is unlikely to be randomized).
3. Changes in transfusion policy recommendations over time make randomization of interventions impractical, suggesting that other study designs may be more appropriate.

The inclusion of NRSIs therefore involved balancing the inclusion of lower-quality studies with the lack of data on the intervention from RCTs. We wanted to include the best available evidence rather than the highest-tier evidence for this high-priority question (involving patient risk reduction, utilization/management of blood, cost, donor involvement). The inclusion of NRSIs increases the risk of systematic differences and confounding across studies, introducing bias that may exaggerate the apparent harm or benefit in one arm.

Types of participants

We included studies of hospitalized adults and children of any age (including preterm neonates), who received at least one allogeneic RBC transfusion for anemia while in hospital. Age was defined according to the center of admission; i.e. preterm infants and neonates were those admitted to the neonatal intensive care unit (NICU) or newborn nursery, children were those admitted to pediatric hospitals (generally birth to 18 years) and adults were those admitted to adult centers. Hospitals included centers that admit inpatients to medical/surgical/oncology wards, intensive care units (ICUs) or other units overnight.

We included only hospitalized patients as they differ from outpatients requiring transfusions. The causes of anemia or transfusion requirements are different for hospitalized patients (severe illness, bleeding versus hematologic disease), and the baseline risks of the outcomes are higher in hospitalized patients. Our objective was therefore to generalize the findings of this review to the target population of hospitalized patients who have characteristics and baseline risks that are different from an outpatient population. If only a subset of patients was eligible for the review within a study, we included it in the descriptive reporting but only included the study in a meta-analysis if we could extract the outcome of the given subset from the total.

Types of interventions

We included studies comparing two volumes of transfusion for a given RBC transfusion event; that is studies comparing a larger volume of transfused RBC units per transfusion (volume/kg/transfusion or volume/transfusion or number of transfusion units) to a smaller volume of RBC units per transfusion or standard practice with respect to the volume of RBC units per transfusion. We defined a transfusion event as a single transfusion, or more than one transfusion in the case of immediate consecutive RBC transfusions. We did not include multiple RBC transfusions given more than six hours apart (for example, during the perioperative period or within an ICU or hospital stay) within the same transfusion event.

For the study intervention, participants should not have been receiving autologous blood transfusion, whole blood or mixed blood components, massive transfusion (≥ 4 units per transfusion event) or blood prime for either cardiopulmonary bypass (CPB), extracorporeal membrane oxygenation (ECMO), or renal replacement therapy (RRT).

The intervention group received a larger volume of RBC units per transfusion (volume/kg/transfusion or volume per transfusion or number of units/transfusion) than the comparative group.

The comparative group(s) received a smaller volume of RBC units per transfusion (volume/kg and volume/transfusion) than the intervention group.

Examples include comparing ≤ 10 mL/kg and > 10 mL/kg RBCs per transfusion in children, or one unit versus two RBC units per transfusion in adults.

Outcome measures

Measuring at least one of the critical or important outcome measures listed below was an inclusion criterion for the review. We contacted authors of studies that measured but did not report one of these outcomes.

Critical outcomes

1. Mortality. We selected 28-day or 30-day hospital mortality first; if this was not available, we used other mortality measures (hospital) after the intervention [32].

Important outcomes

1. Length of hospital stay
2. Hospital-free days (or 28 days hospital-free less length of hospital stay)
3. Transfusion reactions including 1) allergic, 2) non-hemolytic (including febrile non-hemolytic), 3) pulmonary complications (TACO [2, 11], transfusion-associated dyspnea [33], and TRALI [34]), 4) hemolytic and ABO incompatibility, 5) transfusion transmissible infections (TTI) [35], and 6) delayed transfusion reactions (hemolysis and serologic reactions). We extracted dichotomous data for each reaction. However, in the case of rare reporting or few included studies, we grouped these to include ‘any transfusion-associated adverse event (TAAE)’.
4. Organ dysfunction as defined by one of 1) Multiple Organ Dysfunction score or Sequential Organ Failure Assessment (SOFA) score [36]; 2) circulatory shock requiring vasoactive agents or myocardial infarction; 3) respiratory dysfunction or failure, defined as need for mechanical ventilation or as recent partial pressure of oxygen in arterial blood/fraction of inspired oxygen (FiO_2) of < 200 mmHg; 4) hematologic dysfunction, defined as a platelet count of < 100,000/µL, or prothrombin activity of < 50%, evidence of disseminated intravascular coagulation, or arterial or venous thrombosis (including deep vein thrombosis, stroke, and pulmonary embolism); 5) renal dysfunction, defined as a urine output of < 500 mL/day, a serum creatinine level of > 1.9 mg/dL, or dialysis for acute renal failure; or 6) hepatic dysfunction, defined as a serum bilirubin level of > 1.9 mg/dL.
    a. For preterm neonates only, organ dysfunction outcomes included bronchopulmonary dysplasia (BPD), necrotizing enterocolitis (NEC), interventricular hemorrhage (IVH), and retinopathy of prematurity (ROP).
5. Total number of individual allogeneic RBC transfusions received during hospital stay.
6. Rebleeding, defined as a drop in hemoglobin of ≥ 2 g/dL within six hours of the first transfusion, is an important outcome for hospitalized patients, including surgical and trauma patients and those with endothelial dysfunction (sepsis, gastrointestinal bleeding, etc.).
    a. Rebleeding episodes were measured per hospital stay.
7. Allogeneic RBC donor exposure (number of allogeneic RBC donors per patient per hospital stay) is an important hematologic outcome for patients requiring multiple and chronic transfusions, and all potential organ transplant or blood product recipients. Exposure to multiple transfusions from multiple donors increases transfusion-related allogeneic antibodies. This may lead to increased risk of hemolytic transfusion reactions, difficulty finding donor compatibility, and pregnancy complications such as hemolytic disease of the fetus. Future organ recipients may also have difficulty finding donor matches due to compatibility of alloantibodies.
    a. Allogeneic donor exposure was measured per hospital stay as specifically stated in the included studies. Assumptions that each RBC transfusion is a unique donor were not made, given that infants/neonates often receive split RBC packs.

Search methods for identification of studies

Electronic searches

We searched the following databases from inception to 5 August 2025.

1. Evidence-Based Medicine Reviews (EBMR; which includes the Cochrane Central Register of Controlled Trials (CENTRAL))
2. MEDLINE Ovid (1946 to current)
3. Embase Ovid (1974 to current)
4. Web of Science
5. LILACS (Latin American and Caribbean Health Science Information database; 1982 to current)
6. Transfusion Evidence Library

The search strategy was restricted to full texts published in English and French only. A primary search was performed by a librarian of CHU Sainte-Justine (PD), who is familiar with the methodology of systematic reviews [37]. Our search strategy is provided in Supplementary material 1.

Searching other resources

We conducted a manual bibliographic search of references for all included studies and relevant systematic reviews found in the primary search.

Data collection and analysis

Selection of studies

We uploaded the results of the study search into EndNote. Two reviewers (GS, MB) independently assessed the titles and abstracts of records retrieved by the search, with two-person screening (GS, MB, NR) for 35% of studies. We obtained the full texts of studies deemed potentially relevant, and two reviewers (NR, MB, GS) independently assessed the full texts for inclusion in the review. Lists of included studies were compared, and any disagreements were resolved by discussion or consultation with a third review author (JL).

We categorized studies excluded at the full-text stage according to the reason for exclusion, including ineligible study design, setting, population, intervention, or outcomes. The unit of selection was the study, so we chose the first principal reference for inclusion when multiple full-text articles reported on the same study. Study selection and reporting followed PRISMA guidelines [38, 39]. We expressed the rate of agreement (concordance) between review authors for the full-text inclusion of studies using the proportion accuracy (%) and the Kappa score [40][41].

Data extraction and management

Two review authors (GS, NR, MB) independently and in duplicate extracted data from the included studies using a previously piloted data extraction form in Microsoft Excel. Any disagreements were solved by discussion or consultation with a third review author (JL). We extracted data according to the guidelines proposed by Cochrane [42].

We extracted the following data from each study.

1. Study data: first author, year of publication, country
2. Study design: RCT or NRSI, hospital setting, number of sites
3. Study population: age range, diagnostic group, number included, number randomized, number transfused, inclusion and exclusion criteria
4. Intervention(s): larger volume per RBC transfusion event
5. Comparison: smaller volume per RBC transfusion event
6. Outcomes: median or average number of allogeneic RBC transfusions in each group, numbers of deaths in each group, secondary outcome types and definitions with numbers for each group
7. Information related to risk of bias: missing data, imputation
8. Sources of funding and conflicts of interest stated by study authors

We contacted study authors for clarification or additional information where necessary. One review author (NR) entered the data into RevMan software [43], and a second review author checked the data entry (MB).

Risk of bias assessment in included studies

Three review authors (NR, MB, JL) independently assessed risk of bias in the included studies using the Cochrane RoB 2 tool for RCTs [44], and version two of the Risk of Bias in Non-Randomized Studies – of Interventions (ROBINS-I V2) tool for non-randomized studies [45]. We assessed risk of bias for the critical outcome of mortality and the other important outcomes included in the summary of findings tables. We summarized the risk of bias by domain across studies for each outcome, with the overall risk of bias being the least favorable assessment across domains.

For RCTs, we evaluated the effect of assignment to the interventions at baseline (as per intention-to-treat (ITT) where available).

The RoB 2 tool includes the following domains:

1. bias arising from the randomization process;
2. bias due to deviations from intended interventions;
3. bias due to missing outcome data;
4. bias in measurement of the outcome;
5. bias in selection of the reported results.

We then estimated an overall risk of bias for each outcome in each study as follows [46]:

1. low risk of bias: the study is at low risk of bias for all domains for this result;
2. some concerns: the study raised some concerns in at least one domain for this result, but is not at high risk of bias for any domain;
3. high risk of bias: the study is at high risk of bias in at least one domain for the result, or there are some concerns for multiple domains such that our confidence in the results is substantially lowered.

For NRSIs, we employed an analogue to ITT by using the 'start of intervention' in experimental and control groups (RBC transfusion started) as the assessed comparison. The ROBINS-I V2 domains for risk of bias in NRSI include:

1. bias due to confounding;
2. bias in classification of intervention;
3. bias in selection of participants into study;
4. bias due to missing data;
5. bias due to measurement of outcomes;
6. bias in selection of the reported result.

The levels of judgment for the risk of bias are:

1. low risk of bias;
2. moderate risk of bias;
3. serious risk of bias;
4. critical risk of bias;
5. no information.

The overall risk of bias for outcomes was based upon the levels of judgment in each domain (the most serious risk of bias) and how often this arose in multiple domains. A level of bias in any of the domains implies that the overall risk of bias for the outcome is at least this severe; the overall risk of bias may be downgraded if all domains show this level of bias. We excluded studies with critical risk of bias from the pooled meta-analysis.

We piloted the risk of bias assessments on two studies for RCT and two studies for NRSI, to ensure agreement, and used Excel template sheets provided by RoB 2 and ROBINS-I to create tables and figures. Any disagreements were resolved through discussion or consultation with a third review author (JL) if needed. We used the risk of bias assessment to determine study quality. Where possible, we attempted to conduct a sensitivity analysis of studies with low risk of bias.

We illustrated the risk of bias for RCTs using 'traffic light' plots on forest plots, and tables of domains in Supplementary material 5 for RCTs and Supplementary material 8 for NRSIs. The risk of bias assessment informed the GRADE assessment and summary of findings tables [47].

Measures of treatment effect

We measured mortality, transfusion-associated adverse events, and organ dysfunction as dichotomous outcomes, and length of stay, hospital-free days, and number of transfusions as continuous outcomes, each measured at 28 days unless otherwise specified.

We used risk ratios (RR) for dichotomous outcomes and mean differences (MD) for continuous outcomes. We analyzed adults and children separately given the difference in population practice, intervention volumes, and baseline risk of outcomes.

Unit of analysis issues

Given the clinical nature of the review, the unit of analysis was the participant for all included studies. For any included cluster-RCT, cross-over, or multi-arm randomized trials, we planned to reanalyze the results according to the guidance provided in Chapter 23 of the *Cochrane Handbook for Systematic Reviews of Interventions* [48]. We only planned to include cross-over studies in the analysis if the outcome at 28 days could be measured after the first randomized period, and before the cross-over. In multi-arm transfusion studies, we combined arms where possible into large- and small-volume transfusion groups, with the participant as the unit of analysis.

Dealing with missing data

We performed analyses on an intention to treat (ITT) basis. We contacted study authors to obtain or clarify any missing data. If we were unable to obtain the missing data, we reported the data as missing or imputed [42]. If a measure remained incompletely reported despite author contact, we attempted to impute the missing data where possible, as described above. If different effect measures were reported, we attempted to transform to the same effect measure where possible.

Reporting bias assessment

We planned to use a funnel plot to estimate the possibility of non-reporting publication bias if at least 10 studies were included in the meta-analysis [49]. In the event that we had sufficient studies, we would have estimated the statistical significance of the symmetry or asymmetry of the funnel plot using Egger's test [50]

Synthesis methods

We conducted a meta-analysis, illustrated by a forest plot, if studies were homogeneous with regard to clinical characteristics (population, clinical intervention, outcome definition), and methods (study design).

Descriptive results

We synthesized and described all study data in a ‘Characteristics of included studies’ table. We tabulated studies by design (RCT and NRSI), as well as by patient age group (adults versus children/neonates), given that the volume of transfusion in adults is typically in units, whereas in pediatrics it is in mL/kg.

Meta-analysis of numerical data

For categorical (dichotomous) outcomes (mortality, transfusion-associated adverse events, organ dysfunction), we combined data using the inverse variance approach to estimate pooled RRs and their 95% confidence intervals (CIs). A correction factor of 0.5 was attributed to all cells of a given contingency table if it contained one or more zero cells. For continuous outcomes (RBC transfusions, length of stay), we combined data with the same units of measure and estimated pooled MDs using an inverse variance method according to Section 15.5 of the *Cochrane Handbook for Systematic Reviews of Interventions* [51]. If a study provided the median and interquartile range (IQR) rather than the mean and standard deviation (SD) (and authors could not provide the mean and SD), we estimated the mean and SD using the methods described by Wan and colleagues [52], using the median, IQR and sample size according to Section 6.5 of the *Cochrane Handbook for Systematic Reviews of Interventions* [42]. Note that if only the median and range were provided, we did not use the range to estimate the SD, given the lack of robustness in estimating an SD from a range. When data were provided according to two subgroups of interest, we combined them using the formula in Table 6.5a of *the Cochrane Handbook for Systematic Reviews of Interventions* [42]. In the summary of findings tables, for continuous outcomes, we calculated the baseline risk in the lower transfusion volume group using the means and relative weights from the analyses.

Synthesis using other methods

If there were too few studies (≤ 2) to perform a meta-analysis, or there was significant bias in the evidence (missing data), we conducted a narrative synthesis according to Synthesis Without Meta-analysis (SWiM), as described in Chapter 12 of the *Cochrane Handbook for Systematic Reviews of Interventions* [53].

Investigation of heterogeneity and subgroup analysis

Given the inclusion of a clinically diverse population of different age groups and of hospitalized patients, we explored statistical heterogeneity in the results by conducting a Chi^2 test across studies and estimating the I^2 statistic, using the DerSimonian and Laird method to measure statistical heterogeneity [54]. The I^2 statistic estimates the proportion of the variation in the effect measures that is due to heterogeneity across studies rather than sampling error. We interpreted a value of $I^2 > 75\%$ as considerable [54].

We stratified all analyses conducted either in adults or children by study design (RCT and NRSI). For each outcome, we reported meta-analyses using forest plots for each design [53].

Older children (weighing > 20 kg) may often resemble adults in clinical practice and transfusion volume. Therefore, we planned to conduct a subgroup analysis of children (term to < 18 years of age) and preterm neonates (< 37 weeks gestation) in the case of sufficient included studies (at least two per category for RCTs and NRSIs) and conduct an interaction test for subgroup effects.

Other planned subgroups within the adult population were based on those in a previous Cochrane review [25], and included: critically ill patients; trauma patients; cardiac surgery patients; surgical patients (general surgical, orthopedic and other non-cardiac/vascular); myocardial infarction patients; postpartum patients; and cancer and hematologic patients.

Sensitivity analysis

We planned to conduct a sensitivity analysis of studies at low risk of bias only when sufficient studies were included.

Certainty of the evidence assessment

We created summary of findings tables using GRADEpro GDT [55], imported into RevMan [43], following Chapter 14 of the *Cochrane Handbook* [47]. We used GRADE to assess the certainty of the body of evidence according to the five GRADE considerations (risk of bias, consistency of effect, imprecision, indirectness, and publication bias) [56]. The overall risk of bias judgment, assessed using RoB 2 and ROBINS-I, informed the assessment of the first domain (risk of bias); the assessment of heterogeneity informed the second domain (consistency); the CIs were used for the third domain (imprecision); a qualitative assessment of the similarity of the target populations, interventions and outcome measurements across studies informed the fourth domain (indirectness); and a funnel plot informed the fifth domain (publication bias). For RCTs, the certainty of the evidence was started at high and downgraded if there were concerns in any of the above domains. For NRSI, the certainty of evidence was started at low certainty, upgraded for large effect sizes and dose-effect response, and downgraded for plausible confounding. We assessed the certainty of evidence as follows.

- High: we are very confident that the true effect lies close to that of the estimate of the effect.
- Moderate: we are moderately confident in the effect estimate: the true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different.
- Low: our confidence in the effect estimate is limited: the true effect may be substantially different from the estimate of the effect.
- Very low: we have very little confidence in the effect estimate: the true effect is likely to be substantially different from the estimate of effect.

Two review authors (FC, NR) independently performed the GRADE assessment, with any disagreements resolved by a third review author (JL). We used plain language to summarize these findings and make practice recommendations. The summary of findings tables included the critical outcome (mortality) and the first six other important outcomes.

Equity considerations

This review includes a diverse group of patients of all ages and those hospitalized for any condition in order to be as inclusive as possible.

Consumer involvement

Consumers were not involved in this review due to limited resources.

RESULTS

Description of studies

Details of the selection process for, and characteristics of, the included studies are specified below, along with information about interventions and study designs.

Results of the search

The search yielded 33,765 citations across the databases (Supplementary material 1). After removal of duplicates, we screened 12,134 citations and included 107 reports for full-text review (including 18 of the 19 full texts from bibliographic review). Of these, we included 12 studies, reported in 15 publications. The PRISMA flow diagram in Figure 1 details study selection, with six studies identified through the initial search (Avdic 2016 [57]; Bowman 2019 [58]; Chantepie 2023 [59, 60]; Hamm 2021 [61]; Khodabux 2009 [62, 63]; Marco-Ayala 2024 [64]), and six from the bibliographic review of relevant studies (Berger 2011 [65]; Byun 2023 [66]; Chantepie 2017 [67]; Lamarche 2019 [68]; Maitland 2019 [69, 70]; Wong 2005 [71]).

Figure 1. Figure 1. Flow diagram of studies

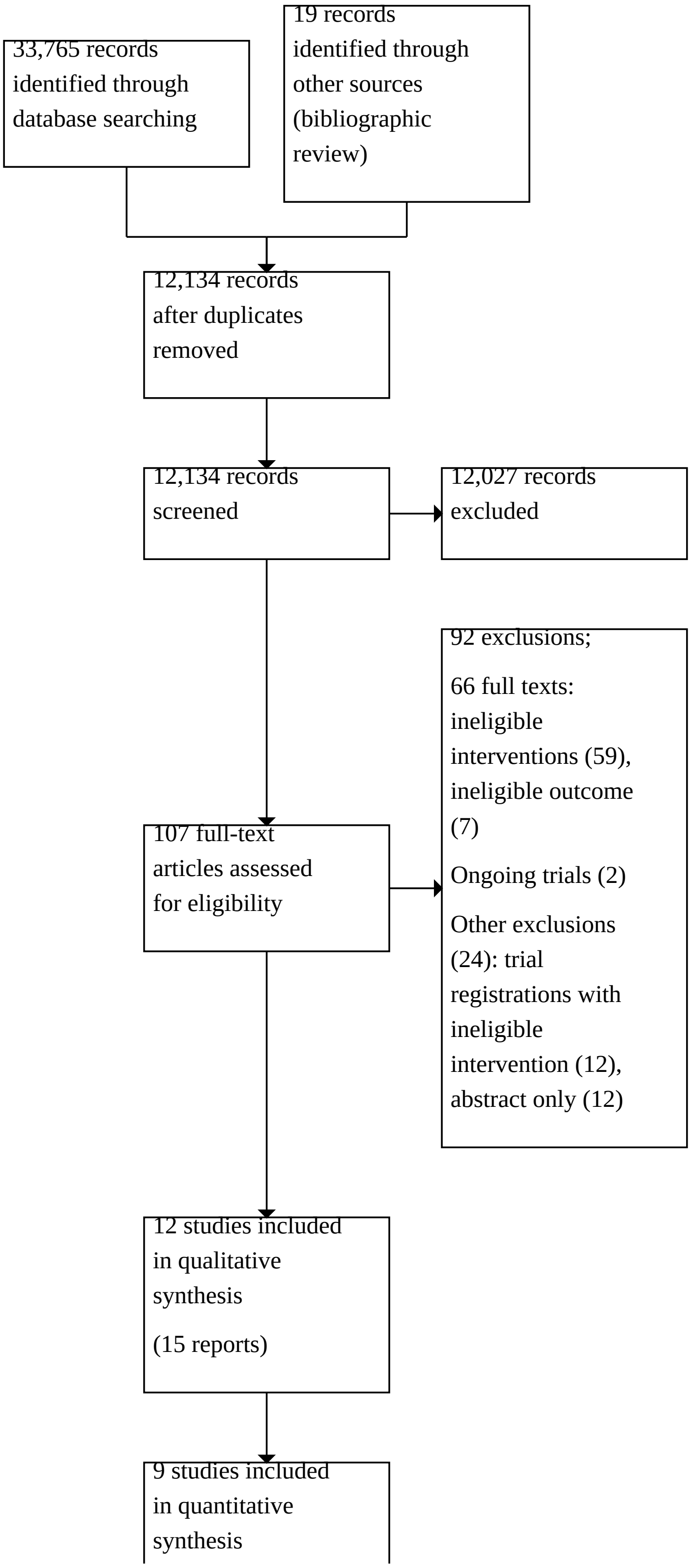
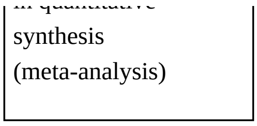


Figure 1. (Continued)



The proportion of agreement between full-text reviews was 95% (agreement 171/total 180) and the Kappa was 0.72, suggesting good agreement between reviewers [41].

On 26 February 2026, we searched all references related to the 12 included studies and found no evidence of retractions or errata in Pubmed or on the editorial websites of the journals that published the included studies.

Included studies

We included a total of 5478 participants (3533 children/neonates and 1945 adults) from 18 sites across 12 studies (two studies from the same site), described in 15 publications. Studies are described in Table 1 and further detailed in Supplementary material 2.

Five studies were randomized controlled trials (RCTs) (Byun 2023; Chantepie 2023; Hamm 2021; Maitland 2019; Wong 2005) and seven were non-randomized studies of interventions (NRSIs) (Avdic 2016; Berger 2011; Bowman 2019; Chantepie 2017; Khodabux 2009; Lamarche 2019; Marco-Ayala 2024). All RCTs randomized participants as the unit of intervention. The NRSIs were all controlled, where the whole inpatient unit/ward or institution, not the individual participant, was the unit of intervention. Six NRSIs were ‘before and after’ studies comparing one unit of RBCs per transfusion to a previous historical cohort of two units of RBCs per transfusion (Table 1, Avdic 2016; Berger 2011; Bowman 2019; Chantepie 2017; Lamarche 2019; Marco-Ayala 2024). In all these studies, administering one unit per RBC transfusion had become the standard practice or institutional policy, supported by standardized order sets within the units. In one parallel-group study conducted in two different centers, two different transfusion strategies were compared, each reflecting the respective center's standard practice (Khodabux 2009). All NRSIs measured outcomes of interest at the participant level.

Studies came from North America, Europe, Asia, and Africa (n = 3 USA, n = 2 Canada, n = 2 France, n = 1 the Netherlands, n = 1 Spain, n = 1 Switzerland, n = 1 Korea, n = 1 Uganda/Malawi). Nine studies included hospitalized adults (Avdic 2016; Berger 2011; Bowman 2019; Byun 2023; Chantepie 2017; Chantepie 2023; Hamm 2021; Lamarche 2019; Marco-Ayala 2024). Three studies included children (Khodabux 2009; Maitland 2019; Wong 2005), of which one study included 3199 children aged two months to 12 years (Maitland 2019), and two studies included a total of 334 preterm neonates (Khodabux 2009; Wong 2005).

Eight studies included adult hematology/oncology patients (Avdic 2016; Berger 2011; Bowman 2019; Byun 2023; Chantepie 2017; Chantepie 2023; Lamarche 2019; Marco-Ayala 2024), and one study included postpartum women (Hamm 2021). One study included children with anemia on the ward (Maitland 2019), and two studies involved preterm infants with anemia in the neonatal intensive care unit (NICU); admitting infants < 32 weeks' gestation (Khodabux 2009), and infants < 1.5 kg requiring transfusion (Wong 2005).

The interventions were consistent across study populations; the nine adult studies compared one unit to two units of packed RBCs (pRBCs) per transfusion event; the pediatric study compared 10 mL/kg to 15 mL/kg of pRBCs (Maitland 2019); and the two neonatal studies compared 15 mL/kg to 20 mL/kg pRBCs per transfusion event (Khodabux 2009; Wong 2005). No other volumes were described by the included studies. Throughout this review, the terms RBCs and pRBCs are used interchangeably, as the intervention was consistent across all included studies.

Excluded studies

We excluded 92 studies from the review at the full-text stage. The reasons for their exclusion are given in the PRISMA flow diagram Figure 1 and described in Supplementary material 3. At the full-text stage, we excluded 59 studies that had ineligible interventions and seven that had ineligible outcomes. There were also 12 trial registrations with ineligible interventions and 12 studies that were only available as abstracts.

There were no studies awaiting classification at the end of the review.

Ongoing Studies

We identified two ongoing studies in pediatrics, the details of which are included in Supplementary material 4. We contacted the studies' authors to find out whether the studies were still ongoing or completed, but received no response.

Risk of bias in included studies

We constructed risk of bias tables, detailing the assessment of studies for each domain by outcome. Summary risk of bias traffic light plots for RCTs are included next to relevant forest plots, and our judgments supporting assessments made with the RoB 2 tool are in Supplementary material 5. Our judgments of risk of bias in NRSIs, using ROBINS-I, are displayed in Supplementary material 8.

Randomized controlled studies

Of the studies included in the meta-analyses, two RCTs reported on mortality; one was at low risk of bias (Chantepie 2023), and one was at high risk of bias (Byun 2023). Two RCTs reported on length of stay, and both were at low risk of bias (Chantepie 2023; Hamm 2021). Three RCTs reported transfusion-associated adverse events; two studies were at low risk of bias (Chantepie 2023; Hamm 2021), and one had a high risk of bias (Byun 2023). For thrombosis, both studies were at low risk of bias (Chantepie 2023; Hamm 2021). For RBC units transfused per patient, one study was at low risk of bias (Hamm 2021), one study had some concerns (Chantepie 2023), and one was at high risk of bias (Byun 2023). For rebleeding, one study was at low risk of bias (Chantepie 2023).

Overall, randomization was well described and groups were balanced in the studies. Studies were all unblinded due to the nature of the intervention. In some cases, we still judged these domains to be at low risk of bias as we did not feel that the lack of blinding would have led to deviations from the interventions. Blinded and objective outcome assessments reduced the risk of detection bias for most outcomes across studies. Despite unblinded studies, many outcomes were still objectively identified and remained at low risk of bias (mortality, length of stay, rebleeding and thrombosis). Incomplete data were low across studies, with only one study having a high attrition rate (Byun 2023). There were no signs of selective reporting across studies.

Non-randomized studies of interventions

Amongst NRSIs, five were included in the meta-analysis on mortality; two were at moderate risk of bias (Chantepie 2017; Marco-Ayala 2024), and three were at serious risk of bias (Berger 2011; Bowman 2019; Lamarche 2019). Four reported on length of stay; one at moderate risk of bias (Marco-Ayala 2024) and three at serious risk of bias (Avdic 2016; Bowman 2019; Lamarche 2019). Three studies reported on transfusion-associated adverse events; one at moderate risk of bias (Berger 2011), and two at serious risk of bias (Bowman 2019; Lamarche 2019). Four studies reported on RBC units transfused per patient stay; two at moderate risk of bias (Chantepie 2017; Marco-Ayala 2024), and two at serious risk of bias (Berger 2011; Bowman 2019). Three reported on rebleeding; one at moderate risk of bias (Chantepie 2017), and two at serious risk of bias (Berger 2011; Lamarche 2019). Overall, most had a serious risk of bias due to concerns about control of confounding, or insufficient adjustment for important confounders when comparing groups with different baseline characteristics across time in before-after studies (Avdic 2016; Berger 2011; Bowman 2019; Chantepie 2017; Lamarche 2019). Deviations from intended interventions and bias in outcome measurements also contributed to the risk of bias. One study was at critical risk of bias due to uncontrolled confounding, and compared two centers with different populations (Khodabux 2009). We excluded the Khodabux 2009 study from the meta-analysis and narrative analysis, as previously planned.

Synthesis of results

Critical outcome

Mortality

Nine studies reported mortality as an outcome (Berger 2011; Bowman 2019; Byun 2023; Chantepie 2023; Khodabux 2009; Lamarche 2019; Maitland 2019; Marco-Ayala 2024; Wong 2005). Mortality was defined in a number of different ways; overall mortality, hospital mortality, 48-hour mortality, 28-day mortality, 30-day mortality, 90-day mortality, 180-day mortality and overall survival.

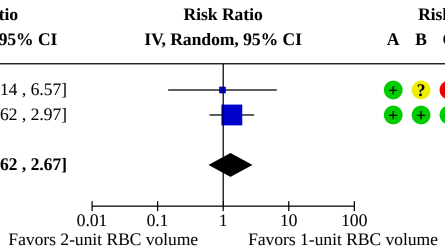
Two RCTs with 322 adult participants showed no difference in mortality (time frame not defined) between the two-unit and one-unit RBC transfusion groups (risk ratio (RR) 1.29, 95% CI 0.62 to 2.67; 2 studies, 322 participants; low-certainty evidence; Figure 2), with no heterogeneity (Chi² = 0.10, df = 1 (P = 0.75); I² = 0%) (Byun 2023; Chantepie 2023). The funnel plots are not presented as there were too few included studies (< 10). The certainty of evidence for the mortality outcome in RCTs was low, as we downgraded the RCT evidence by one level for high risk of bias and one level for imprecision of estimates (Summary of findings 1).

Figure 2. Figure 2. Comparison of 2-unit versus 1-unit RBC transfusion volume (adults) in randomized controlled trials (RCTs) on outcome of 30-day mortality

Study or Subgroup	2-unit RBC volume Events	2-unit RBC volume Total	1-unit RBC volume Events	1-unit RBC volume Total	Weight	Risk Ratio IV, Random, 95% CI	A	B	C	D	E	F
Byun 2023	2	39	2	38	14.5%	0.97 [0.14 , 6.57]	+	?	−	+	+	−
Chantepie 2023	13	120	10	125	85.5%	1.35 [0.62 , 2.97]	+	+	+	+	+	+
Total (Wald[a])		**159**		**163**	**100.0%**	**1.29 [0.62 , 2.67]**						
Total events:	15		12									

Test for overall effect: Z = 0.69 (P = 0.49)

Heterogeneity: Tau² (DL[b]) = 0.00; Chi² = 0.10, df = 1 (P = 0.75); I² = 0%



Footnotes

[a]CI calculated by Wald-type method.

[b]Tau² calculated by DerSimonian and Laird method.

Risk of bias legend

(A) Bias arising from the randomization process
(B) Bias due to deviations from intended interventions
(C) Bias due to missing outcome data
(D) Bias in measurement of the outcome
(E) Bias in selection of the reported result
(F) Overall bias

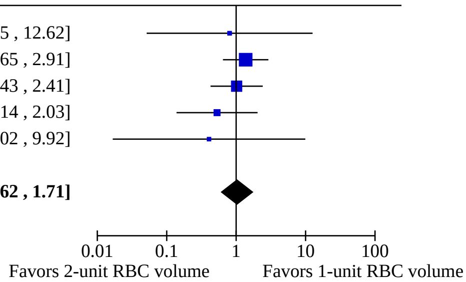
Five NRSI studies reported 30-day mortality in adults (Berger 2011; Bowman 2019; Chantepie 2017; Lamarche 2019; Marco-Ayala 2024). We conducted a meta-analysis of 30-day mortality which included all five NRSIs of adults admitted to malignant hematology wards (1198 participants). We observed no difference in 30-day mortality between the two-unit RBC group and the one-unit RBC group (RR 1.03, 95% CI 0.62 to 1.71; 5 studies, 1198 participants; low-certainty evidence; Figure 3), with no heterogeneity (Chi² = 1.85, df = 4 (P =

0.76); I^2 = 0%). The funnel plots are not presented as there were too few included studies (< 10). The certainty of evidence for the 30-day mortality outcome in NRSIs was low, as we downgraded by one level for serious risk of bias and one level for imprecision of estimates (Summary of findings 2)

Figure 3. Figure 3. Comparison of 2-unit versus 1-unit RBC transfusion volume (adults) in non-randomized studies of interventions (NRSIs) on outcome of 30-day mortality

	2-unit RBC volume		1-unit RBC volume			Risk Ratio	Risk Ratio
Study or Subgroup	Events	Total	Events	Total	Weight	M-H, Random, 95% CI	M-H, Random, 95% CI
Berger 2011	1	77	1	62	3.4%	0.81 [0.05 , 12.62]	
Bowman 2019	16	147	10	126	45.4%	1.37 [0.65 , 2.91]	
Chantepie 2017	12	186	8	126	34.4%	1.02 [0.43 , 2.41]	
Lamarche 2019	3	65	6	69	14.3%	0.53 [0.14 , 2.03]	
Marco-Ayala 2024	0	153	1	187	2.5%	0.41 [0.02 , 9.92]	
Total (Wald[a])		**628**		**570**	**100.0%**	**1.03 [0.62 , 1.71]**	
Total events:	32		26				

Test for overall effect: Z = 0.11 (P = 0.91)



Heterogeneity: Tau² (DL[b]) = 0.00; Chi² = 1.85, df = 4 (P = 0.76); I² = 0%

Footnotes

[a]CI calculated by Wald-type method.

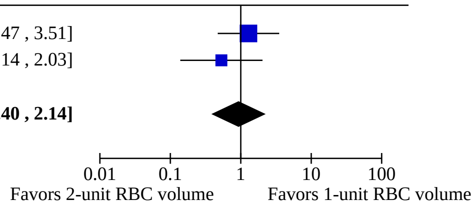
[b]Tau² calculated by DerSimonian and Laird method.

Two of the adult studies above also reported hospital mortality (2 NRSIs, 407 participants) (Bowman 2019; Lamarche 2019). We conducted a meta-analysis of the two NRSIs' hospital mortality results, given that both studies compared the same intervention in malignant hematology patients (Bowman 2019; Lamarche 2019). We observed no difference in risk of hospital mortality between the two-unit RBC group and the one-unit RBC group (RR 0.93, 95% CI 0.40 to 2.14; 2 studies, 407 participants; very low-certainty evidence; Figure 4), with low heterogeneity (Chi^2 = 1.07, df = 1 (P = 0.30); I^2 = 7%). Both of the studies reporting hospital mortality also reported results for 30-day mortality and were included in the larger meta-analysis for that outcome. Therefore, they are not included in the summary of findings table, but rather reported as a secondary analysis. The funnel plots are not presented for hospital mortality as there were too few included studies (< 10).

Figure 4. Figure 4. Comparison of 2-unit versus 1-unit RBC transfusion volume (adults) in non-randomized studies of interventions (NRSIs) on outcome of hospital mortality

	2-unit RBC volume		1-unit RBC volume			Risk Ratio	Risk Ratio
Study or Subgroup	Events	Total	Events	Total	Weight	IV, Random, 95% CI	IV, Random, 95% CI
Bowman 2019	9	147	6	126	63.2%	1.29 [0.47 , 3.51]	
Lamarche 2019	3	65	6	69	36.8%	0.53 [0.14 , 2.03]	
Total (Wald[a])		**212**		**195**	**100.0%**	**0.93 [0.40 , 2.14]**	
Total events:	12		12				

Test for overall effect: Z = 0.17 (P = 0.86)



Heterogeneity: Tau² (DL[b]) = 0.02; Chi² = 1.07, df = 1 (P = 0.30); I² = 6%

Footnotes

[a]CI calculated by Wald-type method.

[b]Tau² calculated by DerSimonian and Laird method.

Three studies in pediatrics reported on mortality (Khodabux 2009; Maitland 2019; Wong 2005), but one NRSI was at critical risk of bias (Khodabux 2009). The two pediatric RCTs reporting mortality had separate interventions, populations, and sample sizes, so heterogeneity precluded meta-analysis or SWiM (Maitland 2019; Wong 2005). One pediatric pilot RCT included 20 preterm neonates and described one death in each of the 20 mL/kg and 15 mL/kg blood transfusion groups (no effect estimates measured) (Wong 2005). The other pediatric RCT included 3196 children and showed no difference in 28-day mortality between the 15 mL/kg and 10 mL/kg blood transfusion groups (55 vs 72; hazard ratio (HR) 0.76, 95% CI 0.54 to 1.08; 3196 participants) (Maitland 2019).

Important outcomes

Length of hospital stay

Nine studies included length of hospital stay as an outcome (Avdic 2016; Bowman 2019; Byun 2023; Chantepie 2017; Chantepie 2023; Hamm 2021; Khodabux 2009; Lamarche 2019; Marco-Ayala 2024). Five studies reported the length of stay as the median and IQR, so we imputed the mean and SD (Bowman 2019; Chantepie 2023; Marco-Ayala 2024). For two of these studies, we combined the imputed mean and SD for the length of stay in two subgroups, to give an overall mean (SD) length of stay (Avdic 2016, allogeneic and autologous groups combined; Lamarche 2019, leukemia and autologous stem cell transplant combined (ASCT)). One study reported length of stay with median and range (Byun 2023). Another study reported the length of stay across three subgroups and over multiple hospital stays (Chantepie 2017). Therefore, we did not include these two studies in the meta-analysis.

We conducted meta-analyses of length of stay with six of the adult studies; one meta-analysis with two RCTs and one meta-analysis with four NRSIs. The meta-analysis of RCTs found no difference in hospital length of stay between the two-unit RBC group and the one-unit RBC group (mean difference (MD) 0.08, 95% CI -0.66 to 0.82; 2 studies, 311 participants; low-certainty evidence; Figure 5), with low heterogeneity ($Chi^2 = 0.52$, df = 1 (P = 0.47); $I^2 = 0\%$) (Chantepie 2023; Hamm 2021). The certainty of evidence for this outcome was low; although the RCTs were at low risk of bias, they provided indirect evidence, and we downgraded by one level for imputation of the mean and one level for imprecision (Summary of findings 1). The meta-analysis of NRSIs found no difference in hospital length of stay between the two-unit RBC group and the one-unit RBC group (MD -0.15, 95% CI-1.68 to 1.38; 4 studies, 1048 participants; very low-certainty evidence); Figure 6) with low heterogeneity ($Chi^2 = 3.71$, df = 3 (P = 0.29); $I^2 = 19\%$) (Avdic 2016; Bowman 2019; Lamarche 2019; Marco-Ayala 2024). The certainty of evidence from the NRSI was very low because we downgraded the NRSI evidence by one level each for serious risk of bias, indirect evidence (imputation) and imprecision (Summary of findings 2). The funnel plots are not presented for the length of hospital stay outcome as there were too few included studies (< 10).

Figure 5. Figure 5. Comparison of 2-unit versus 1-unit RBC transfusion volume (adults) in randomized controlled trials (RCTs) on outcome of length of hospital stay

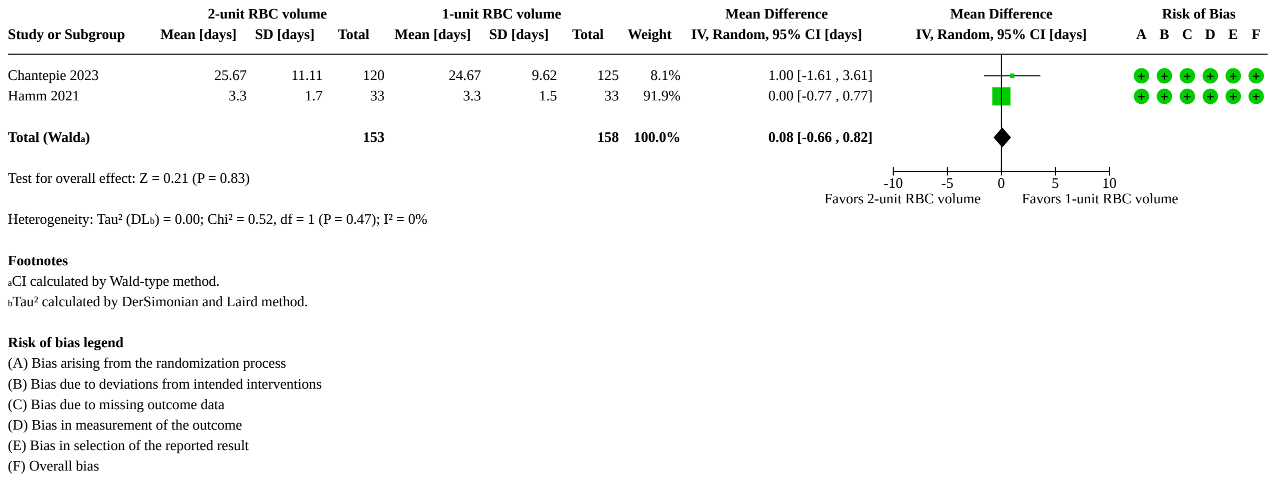
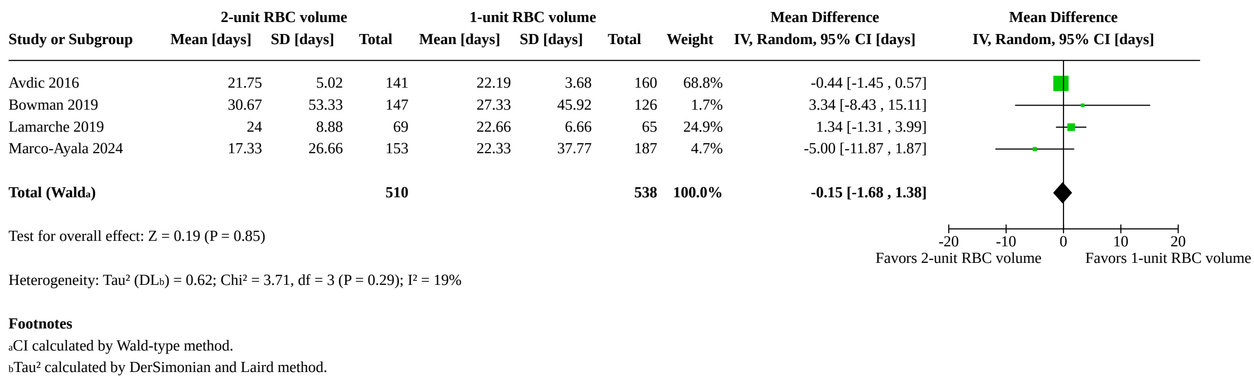


Figure 6. Figure 6. Comparison of 2-unit versus 1-unit RBC transfusion volume (adults) in non-randomized studies of interventions (NRSIs) on outcome of length of hospital stay



Only one pediatric study reported length of stay (Khodabux 2009), but the study was at critical risk of bias, so we have not reported its results here.

Hospital-free days

None of the included studies reported hospital-free days, nor could we calculate hospital-free days based on data provided in any included studies.

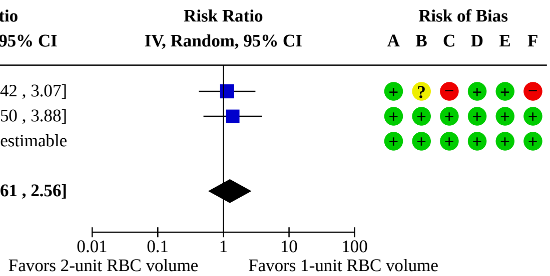
Transfusion-associated adverse events (TAAE)

Eight studies reported on transfusion-associated adverse events (TAAE) (Berger 2011; Bowman 2019; Byun 2023; Chantepie 2023; Hamm 2021; Lamarche 2019; Maitland 2019; Wong 2005), two of which were in children. All eight studies defined the TAAEs similarly as transfusion complications, transfusion reactions or transfusion adverse events (febrile reactions and pulmonary edema). In addition to TAAE, one study further reported allergic transfusion events (Maitland 2019), and two reported transfusion-related acute lung injury (TRALI) (Maitland 2019; Wong 2005).

We conducted meta-analyses of TAAE with all six of the adult studies; one with three RCTs (Byun 2023; Chantepie 2023; Hamm 2021), and one with three NRSI (Bowman 2019; Lamarche 2019). The meta-analysis of RCTs found no difference in TAAEs between the two-unit RBC transfusion group and the one-unit RBC transfusion group (RR 1.25, 95% CI 0.61 to 2.56; 3 studies, 388 participants, low-certainty evidence; Figure 7) with low heterogeneity ($Chi^2 = 0.08$, df = 1 (P = 0.78); $I^2 = 0\%$). The certainty of evidence for TAAE in randomized studies is low as one of the RCTs was at high risk of bias, and we further downgraded for indirectness of populations (Summary of findings 1). The meta-analysis of NRSIs found no difference in TAAEs between the two-unit RBC transfusion group and the one-unit RBC transfusion group (RR 0.74, 95% CI 0.30 to 1.82; 3 studies, 546 participants, very low-certainty evidence; Figure 8) with low heterogeneity ($Chi^2 = 0.59$, df = 2 (P = 0.74); $I^2 = 0\%$). The certainty of evidence for TAAE in NRSI was very low as we downgraded the NRSI by one level for serious risk of bias and two levels for imprecision (Summary of findings 2). The funnel plot for TAAE is not presented as there were too few included studies (< 10).

Figure 7. Figure 7. Comparison of 2-unit versus 1-unit RBC transfusion volume (adults) in randomized controlled trials (RCTs) on outcome of transfusion associated adverse event

Study or Subgroup	2-unit RBC volume Events	2-unit RBC volume Total	1-unit RBC volume Events	1-unit RBC volume Total	Weight	Risk Ratio IV, Random, 95% CI
Byun 2023	7	39	6	38	51.7%	1.14 [0.42 , 3.07]
Chantepie 2023	8	120	6	125	48.3%	1.39 [0.50 , 3.88]
Hamm 2021	0	33	0	33		Not estimable
Total (Wald[a])		**192**		**196**	**100.0%**	**1.25 [0.61 , 2.56]**
Total events:	15		12			
Test for overall effect: Z = 0.62 (P = 0.54)						



Heterogeneity: Tau^2 (DL_b) = 0.00; $Chi^2 = 0.08$, df = 1 (P = 0.78); $I^2 = 0\%$

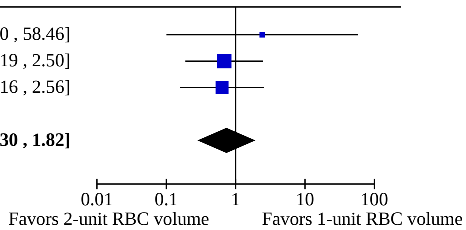
Footnotes
[a]CI calculated by Wald-type method.
[b]Tau^2 calculated by DerSimonian and Laird method.

Risk of bias legend
(A) Bias arising from the randomization process
(B) Bias due to deviations from intended interventions
(C) Bias due to missing outcome data
(D) Bias in measurement of the outcome
(E) Bias in selection of the reported result
(F) Overall bias

Figure 8. Figure 8. Comparison of 2-unit versus 1-unit RBC transfusion volume (adults) in non-randomized studies of interventions (NRSIs) on outcome of transfusion associated adverse event

Study or Subgroup	2-unit RBC volume Events	2-unit RBC volume Total	1-unit RBC volume Events	1-unit RBC volume Total	Weight	Risk Ratio M-H, Random, 95% CI
Berger 2011	1	77	0	62	8.1%	2.42 [0.10 , 58.46]
Bowman 2019	4	147	5	126	49.3%	0.69 [0.19 , 2.50]
Lamarche 2019	3	65	5	69	42.6%	0.64 [0.16 , 2.56]
Total (Wald[a])		**289**		**257**	**100.0%**	**0.74 [0.30 , 1.82]**
Total events:	8		10			

Test for overall effect: Z = 0.66 (P = 0.51)



Heterogeneity: Tau² (DL[b]) = 0.00; Chi² = 0.59, df = 2 (P = 0.74); I² = 0%

Footnotes

[a]CI calculated by Wald-type method.

[b]Tau² calculated by DerSimonian and Laird method.

The two pediatric studies that described TAAE were not compared given the heterogeneity in the populations and intervention volumes. Maitland 2019 described 25/1598 suspected allergic reactions in the larger RBC volume group (15 mL/kg) and 20/1598 suspected allergic reactions in the smaller RBC volume group (10 mL/kg) (P = 0.55). Maitland 2019 also described two transfusion-associated lung injury events in the larger RBC volume group (15 mL/kg) and three events in the smaller RBC volume group (10 mL/kg)(P = 1.00). Wong 2005 had no TAAE in either group.

Organ dysfunction

The only organ dysfunction described in the included adult studies was hematological dysfunction, defined as thrombosis, which is included in the summary of findings table. Otherwise, the only pediatric organ dysfunction was retinopathy in preterm infants.

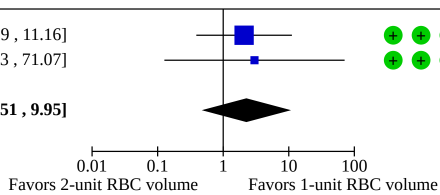
Thrombosis

Two adult RCTs reported on thrombotic events (Chantepie 2023; Hamm 2021). We conducted a meta-analysis and found no difference in thrombotic events between the two-unit RBC transfusion group and the one-unit RBC transfusion group (RR 2.26, 95% CI 0.51 to 9.95; 2 studies, 311 participants, very low-certainty evidence; Figure 9) with low heterogeneity (Chi² = 0.04, df = 1 (P = 0.84); I² = 0%). We downgraded the certainty of evidence for the outcome thrombosis by one level due to indirectness and two levels for imprecision (Summary of findings 1). The funnel plot is not presented as there were too few included studies (< 10).

Figure 9. Figure 9. Comparison of 2-unit versus 1-unit RBC transfusion volume (adults) in randomized controlled trials (RCTs) on outcome of thrombotic event

Study or Subgroup	2-unit RBC volume Events	2-unit RBC volume Total	1-unit RBC volume Events	1-unit RBC volume Total	Weight	Risk Ratio IV, Random, 95% CI	Risk of Bias A	B	C	D	E	F
Chantepie 2023	4	120	2	125	78.0%	2.08 [0.39 , 11.16]	+	+	+	+	+	+
Hamm 2021	1	33	0	33	22.0%	3.00 [0.13 , 71.07]	+	+	+	+	+	+
Total (Wald[a])		**153**		**158**	**100.0%**	**2.26 [0.51 , 9.95]**						
Total events:	5		2									

Test for overall effect: Z = 1.08 (P = 0.28)



Heterogeneity: Tau² (DL[b], 95% CI) = 0.00 [0.00 , 66.03]; Chi² = 0.04, df = 1 (P = 0.84); I² = 0%

Footnotes

[a]CI calculated by Wald-type method.

[b]Tau² calculated by DerSimonian and Laird method.

Risk of bias legend

(A) Bias arising from the randomization process

(B) Bias due to deviations from intended interventions

(C) Bias due to missing outcome data

(D) Bias in measurement of the outcome

(E) Bias in selection of the reported result

(F) Overall bias

Retinopathy

In pediatric studies, Wong 2005 reported retinopathy of prematurity (ROP) in their RCT (20 participants), with 2/10 cases in both transfusion arms.

Total number of red blood cell transfusions during hospital stay

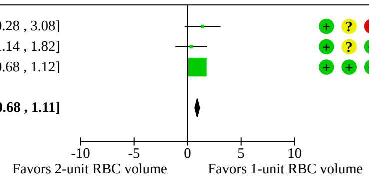
Total RBC volume (in units) per patient per admission

All nine adult studies reported the number of RBC units transfused per patient per admission. One study expressed this outcome as the median and range (Lamarche 2019), and another study reported only the subgroup mean without SD or 95% CI (Avdic 2016), precluding inclusion of either study in the meta-analysis.

We conducted meta-analyses of the number of RBC units transfused per patient per admission in each group, with the seven remaining adult studies (three RCTs, four NRSIs). In the meta-analysis of RCTs, we found no difference between the total number of RBC units transfused during the hospital stay between the two-unit RBC transfusion group and the one-unit RBC transfusion group (MD 0.90 units, 95% CI 0.68 to 1.11; 3 studies, 378 participants; low-certainty evidence; Figure 10), with low heterogeneity ($Chi^2 = 0.89$ df = 2 ($P = 0.64$); $I^2 = 0\%$) (Byun 2023; Chantepie 2023; Hamm 2021). The certainty of evidence was low as we downgraded by one level for high risk of bias and one level for indirectness (Summary of findings 1).

Figure 10. Figure 10. Comparison of 2-unit versus 1-unit RBC transfusion volume (adults) in randomized controlled trials (RCTs) on outcome of RBC units transfused per patient admission

	2-unit RBC volume			1-unit RBC volume				Mean Difference	Risk of Bias					
Study or Subgroup	Mean [units]	SD [units]	Total	Mean [units]	SD [units]	Total	Weight	IV, Random, 95% CI [units]	A	B	C	D	E	F
Byun 2023	5.3	4.2	39	3.9	2.8	28	1.6%	1.40 [-0.28 , 3.08]	+	?	−	+	+	−
Chantepie 2023	8	5.92	120	7.66	5.92	125	2.1%	0.34 [-1.14 , 1.82]	+	?	+	+	+	?
Hamm 2021	2.1	0.4	33	1.2	0.5	33	96.3%	0.90 [0.68 , 1.12]	+	+	+	+	+	+
Total (Wald[a])			**192**			**186**	**100.0%**	**0.90 [0.68 , 1.11]**						



Test for overall effect: Z = 8.20 (P < 0.00001)

Heterogeneity: Tau² (DL[b]) = 0.00; Chi² = 0.89, df = 2 (P = 0.64); I² = 0%

Footnotes

[a]CI calculated by Wald-type method.

[b]Tau² calculated by DerSimonian and Laird method.

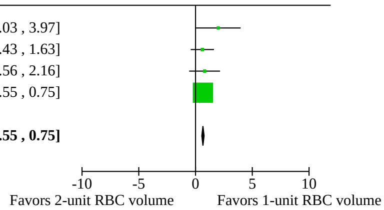
Risk of bias legend

(A) Bias arising from the randomization process
(B) Bias due to deviations from intended interventions
(C) Bias due to missing outcome data
(D) Bias in measurement of the outcome
(E) Bias in selection of the reported result
(F) Overall bias

In the meta-analysis of NRSIs, the two-unit RBC transfusion group received on average 0.65 more units of RBC during their hospital stay compared to the one-unit RBC transfusion group (MD 0.65 units, 95% CI 0.55 to 0.75; 4 studies, 1064 participants; low-certainty evidence; Figure 11), with low heterogeneity ($Chi^2 = 1.85$ df = 3 ($P = 0.60$); $I^2 = 0\%$) (Berger 2011; Bowman 2019; Chantepie 2017; Marco-Ayala 2024). The certainty of evidence for this outcome was low, as we downgraded by one level for serious risk of bias and one level for indirect evidence (imputation of mean). The funnel plots are not presented as there were too few included studies (< 10).

Figure 11. Figure 11. Comparison of 2-unit versus 1-unit RBC transfusion volume (adults) in non-randomized studies of interventions (NRSIs) on outcome of RBC units transfused per patient admission

	2-unit RBC volume			1-unit RBC volume				Mean Difference
Study or Subgroup	Mean [units]	SD [units]	Total	Mean [units]	SD [units]	Total	Weight	IV, Random, 95% CI [units]
Berger 2011	8.33	6.67	77	6.33	5.18	62	0.3%	2.00 [0.03 , 3.97]
Bowman 2019	5.1	4.2	147	4.5	4.4	126	1.0%	0.60 [-0.43 , 1.63]
Chantepie 2017	8.1	6.2	186	7.3	5.9	126	0.5%	0.80 [-0.56 , 2.16]
Marco-Ayala 2024	2.07	0.57	153	1.42	0.32	187	98.2%	0.65 [0.55 , 0.75]
Total (Wald[a])			**563**			**501**	**100.0%**	**0.65 [0.55 , 0.75]**



Test for overall effect: Z = 12.76 (P < 0.00001)

Heterogeneity: Tau² (DL[b]) = 0.00; Chi² = 1.85, df = 3 (P = 0.60); I² = 0%

Footnotes

[a]CI calculated by Wald-type method.

[b]Tau² calculated by DerSimonian and Laird method.

RBC volume per patient

Only one pediatric study reported the volume of RBCs transfused per patient per group (Khodabux 2009), but this study was at critical risk of bias.

RBC units per transfusion event

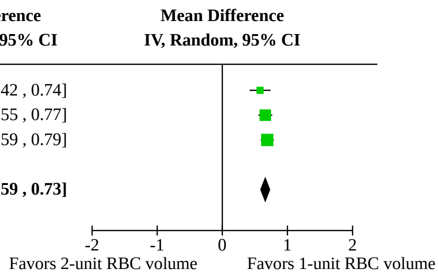
Five adult studies (two RCTs and three NRSIs), all in malignant hematology patients, reported on the number of RBC units transfused per transfusion event (Berger 2011; Byun 2023; Chantepie 2017; Chantepie 2023; Marco-Ayala 2024). Transfusion events are the number of distinct transfusion decisions, or transfusion episodes. This outcome measures the adherence to the allocation strategy by measuring the exact volume, in units, that each group receives per transfusion event.

One RCT reported this outcome as median and range, so we did not attempt imputation (Byun 2023). The other RCT reported a mean (SD) of 1.83 (0.57) units transfused per event in the two-unit RBC group and 1.09 (0.31) units transfused per event in the one-unit RBC group (P < 0.001) (Chantepie 2023).

We conducted a meta-analysis of the number of RBC units per transfusion event per patient in the three NRSIs (791 participants) (Berger 2011; Chantepie 2017; Marco-Ayala 2024). The meta-analysis demonstrated that the two-unit RBC transfusion group received, on average, 0.66 more units per transfusion event than the one-unit RBC transfusion group (MD) 0.66 units, 95% CI 0.59 to 0.73; 3 studies, 791 participants; Figure 12), with low heterogeneity (Chi² = 1.29, df = 2 (P = 0.52); I² = 0%). The funnel plots are not presented as there were too few included studies (< 10). This outcome is not presented in the summary of findings table, as it is a secondary outcome.

Figure 12. Figure 12. Comparison of 2-unit versus 1-unit RBC transfusion volume (adults) in non-randomized studies of interventions (NRSIs) on outcome of RBC units per transfusion episode

	2-unit RBC volume			1-unit RBC volume				Mean Difference
Study or Subgroup	Mean	SD	Total	Mean	SD	Total	Weight	IV, Random, 95% CI
Berger 2011	1.76	0.49	77	1.18	0.47	62	17.3%	0.58 [0.42 , 0.74]
Chantepie 2017	1.66	0.74	186	1	0.01	126	39.3%	0.66 [0.55 , 0.77]
Marco-Ayala 2024	1.85	0.57	153	1.16	0.32	187	43.3%	0.69 [0.59 , 0.79]
Total (Wald[a])			**416**			**375**	**100.0%**	**0.66 [0.59 , 0.73]**



Test for overall effect: Z = 19.37 (P < 0.00001)

Heterogeneity: Tau² (DL[b]) = 0.00; Chi² = 1.29, df = 2 (P = 0.52); I² = 0%

Footnotes

[a]CI calculated by Wald-type method.

[b]Tau² calculated by DerSimonian and Laird method.

RBC transfusion events per patient

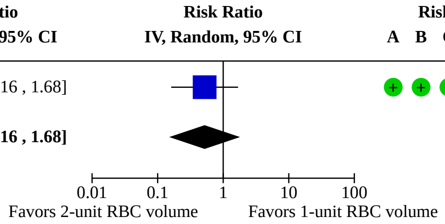
Only one study, an NRSI in adult malignant hematology patients, reported on the number of RBC transfusion events per patient (Bowman 2019). The study number was insufficient to conduct a meta-analysis. The study reported that patients in the two-unit RBC transfusion group had a mean of 2.7 (SD 2.0) transfusion events per stay, compared to a mean of 4.1 (SD 4.0) RBC transfusion events in the one-unit RBC transfusion group (Bowman 2019).

Rebleeding

Four studies (one RCT and three NRSIs) reported rebleeding as an outcome (Berger 2011; Chantepie 2017; Chantepie 2023; Lamarche 2019). All were adult studies of patients admitted for malignant hematological conditions. All studies defined rebleeding as Grade 3 or more, suggesting major bleeding/hemorrhage requiring RBC transfusion over baseline transfusion needs. One RCT found no difference in rebleeding events between the one-unit RBC group and the two-unit RBC transfusion group (RR 0.52, 95% CI 0.16 to 1.68; 1 study; 245 participants, moderate-certainty evidence;Figure 13) (Summary of findings 1 Chantepie 2023). The certainty of evidence was moderate as we downgraded by one level for imprecision (Summary of findings 1). We conducted a meta-analysis of rebleeding events with all three NRSI studies (Berger 2011; Chantepie 2017; Lamarche 2019). There was no difference between rebleeding events between the two-unit RBC transfusion group and the one-unit RBC transfusion group (RR 0.64, 95% CI 0.36 to 1.14; 3 NRSI; 585 participants; very low-certainty evidence; Figure 14), with low heterogeneity (Chi² = 2.19, df = 2 (P = 0.33); I² = 9%). The certainty of evidence for this outcome was very low as we downgraded the NRSIs by two levels for serious risk of bias and one level for imprecision (Summary of findings 2). The funnel plot is not presented as there were too few included studies (< 10).

Figure 13. Figure 13. Comparison of 2-unit versus 1-unit RBC transfusion volume (adults) in randomized controlled trials (RCTs) on outcome of rebleeding

	2-unit RBC volume		1-unit RBC volume			Risk Ratio	Risk of Bias					
Study or Subgroup	Events	Total	Events	Total	Weight	IV, Random, 95% CI	A	B	C	D	E	F
Chantepie 2023	4	120	8	125	100.0%	0.52 [0.16 , 1.68]	+	+	+	+	+	+
Total		**120**		**125**	**100.0%**	**0.52 [0.16 , 1.68]**						
Total events:	4		8									
Test for overall effect: Z = 1.09 (P = 0.28)												

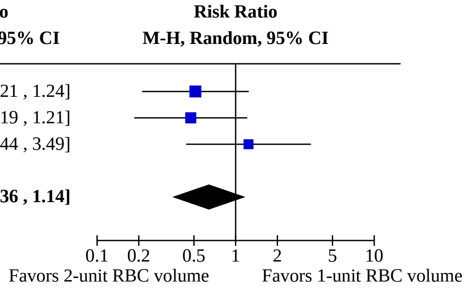


Heterogeneity: Not applicable

Risk of bias legend
(A) Bias arising from the randomization process
(B) Bias due to deviations from intended interventions
(C) Bias due to missing outcome data
(D) Bias in measurement of the outcome
(E) Bias in selection of the reported result
(F) Overall bias

Figure 14. Figure 14. Comparison of 2-unit versus 1-unit RBC transfusion volume (adults) in non-randomized studies of interventions (NRSIs) on outcome of rebleeding

	2-unit RBC volume		1-unit RBC volume			Risk Ratio
Study or Subgroup	Events	Total	Events	Total	Weight	M-H, Random, 95% CI
Berger 2011	7	77	11	62	37.7%	0.51 [0.21 , 1.24]
Chantepie 2017	7	186	10	126	34.0%	0.47 [0.19 , 1.21]
Lamarche 2019	7	65	6	69	28.3%	1.24 [0.44 , 3.49]
Total (Wald[a])		**328**		**257**	**100.0%**	**0.64 [0.36 , 1.14]**
Total events:	21		27			
Test for overall effect: Z = 1.52 (P = 0.13)						



Heterogeneity: Tau² (DL[b]) = 0.02; Chi² = 2.19, df = 2 (P = 0.33); I² = 9%

Footnotes
[a]CI calculated by Wald-type method.
[b]Tau² calculated by DerSimonian and Laird method.

Allogeneic RBC donor exposure

No studies reported on allogeneic RBC donor exposure between patients in the one-unit RBC transfusion group and the two-unit RBC transfusion volume group.

Subgroup analyses

There were insufficient studies to conduct planned subgroup analyses. In children, the three included studies were too heterogenous in population and intervention (volume of RBC transfusion) to conduct a subgroup analysis.

A summary of all analyses is available in Supplementary material 6, along with a data package in Supplementary material 7.

Equity assessment

With a vision of being fully inclusive, and understanding that populations are underrepresented in research, this review includes a broad population of hospitalized patients, including patients of all ages, women (including pregnant women) and children (including preterm infants). Databases searched were broad in range. In addition to Embase and MEDLINE, we included the Web of Science, LILACs, and Transfusion Evidence Reviews.

The intervention of RBC transfusion is broadly available in all countries, although it may be more difficult to access in lower-resource settings such as in low- and middle-income countries (LMIC). The research question is potentially more important in low-resource settings, where limiting the volume of RBC used at each transfusion event may result in a more efficient and equitable use of limited blood resources. This review includes only one study conducted in a LMIC; a large pediatric RCT conducted in Uganda/Malawi (Maitland 2019). All other studies were conducted in high-income countries.

The outcomes assessed in the review include important equity outcomes and outcomes relevant for patients, practitioners, payers, and policies, such as mortality, length of hospital stay, organ dysfunction and RBC utilization. Length of hospital stay and organ dysfunction are not only patient-centered outcomes, but are relevant for practitioners, hospital stakeholders (bed flow), payers (cost), and hospital administrators. Outcomes related to blood management and transfusion events are important to consider for management of blood services in all hospitals and countries.

Reporting biases

Funnel plots were not used to examine publication bias as there were too few studies included in each meta-analysis (all below 10). We cannot report on publication bias in this review.

DISCUSSION

Given the inherent risks of red blood cell (RBC) transfusions, and the limited availability of this resource, recommendations have been made to transfuse one unit of RBC at a time in adults. This is based on sparse evidence. Given the paucity of data, no recommendations have been made for pediatric patients. This review aimed to fill that knowledge gap.

Summary of main results

In adults, when comparing two units of RBCs (larger volume) to one unit of RBCs (smaller volume) for a single transfusion event, there was no difference in 30-day mortality. The findings are based on the inclusion of two RCTS and five NRSI at high or serious risk of bias, respectively. Our confidence in the certainty of evidence is low for RCTs and NRSIs reporting this outcome. The limited number, heterogeneity, and high risk of bias of existing studies in children preclude any firm conclusions about mortality risk between larger-volume and smaller-volume groups in this population.

As for other important outcomes, there was no evidence of any differences in hospital length of stay, transfusion-associated adverse event, thrombosis or rebleeding when comparing adult patients transfused using two units of RBCs rather than one unit. Although no differences were observed in the aforementioned outcomes, the two-unit transfusion group was associated with a higher number of RBC units transfused per event (consistent with adherence to the intervention). However, the RCT evidence did not find a reduction in the total number of RBC units transfused during the hospitalization, and more units were received in the two-unit group in the NRSIs).

Our confidence in the certainty of evidence for length of stay and transfusion-associated adverse events is low for RCTs and very low for NRSIs. No NRSIs assessed thrombosis, and our confidence in the certainty of evidence for thrombosis from RCTs was very low. For both RCTs and NRSIs, our confidence in the certainty of evidence is low for blood received during the hospital stay. For rebleeding, our confidence is moderate for RCTs and very low for NRSIs.

The results indicate that a restrictive transfusion strategy, involving lower transfusion volumes per transfusion event, is not associated with adverse clinical outcomes, based on the low- or very low-certainty evidence available. At the same time, it may lead to a reduction in the total amount of blood administered, highlighting potential benefits in resource conservation and blood product utilization with this strategy.

Limitations of the evidence included in the review

This review has several limitations. The literature search was restricted to studies published in English or French, and therefore studies in other languages may have been missed. While this may result in possible language bias, some recent studies have suggested that there were no major differences in summary estimates of meta-analyses restricted to the English language, compared to those with no language restrictions [72]. We excluded conference abstracts; however, we contacted the corresponding authors to determine whether their findings had been subsequently published in full manuscripts.

Most importantly, the review results are limited by the number of included studies and underlying quality of the evidence. Few studies met the inclusion criteria, and these studies involved heterogenous groups of hospitalized patients. The evidence is also of low certainty. Two of the five included RCTs had some concerns of bias or a high risk of bias, inherently limiting the quality of the pooled analyses and the final certainty of the evidence. The risk of bias was often due to lack of information on randomization and missing data, which may have affected outcome assessment or important co-interventions. The included NRSIs were also at serious risk of bias, mostly due to deviation in confounding related to study design and temporal confounding across comparison groups.

For each outcome, few studies and few patients were available, limiting the precision of our results. In some reports, it was not clearly specified whether the reported transfusion volume (expressed as number of units or mL/kg per transfusion) referred to the prescribed or the actual volume administered to patients. In addition, detailed descriptions of the RBC unit volumes were often lacking in the reviewed literature. These limitations suggest ill-defined interventions. Combined with variable baseline risks across studies for several outcomes, these limitations suggest some degree of indirectness in our results, reflecting variations in the target populations and analyzed interventions. In addition, data imputation — from medians with interquartile ranges to means with standard deviations — was necessary when outcome units were not clearly reported by the authors, further contributing to the imprecision of our findings. Taken together, these limitations contributed to downgrading the certainty of the evidence to low or very low for most outcomes.

Finally, the review included only hospitalized patients (with the exclusion of people receiving extracorporeal membrane oxygenation (ECMO) and hemofiltration circuits priming), limiting the generalizability of its findings to individuals transfused in outpatient or other non-hospital settings.

Limitations of the review processes

It is very unlikely that we missed the inclusion of relevant studies comparing two strategies of blood volumes in hospitalized patients, given the extensive search strategies designed for and run on multiple electronic databases and clinical trial registries (Supplementary material 1). We also extensively reviewed bibliographic references and other systematic reviews that allowed us to find supplemental entries [73].

We contacted multiple authors to confirm findings and data, but received only two responses (Chantepie 2023; Hamm 2021), leaving us with incomplete data. The number of papers on pediatric or neonatal patients was too small and too heterogenous to undertake a quantitative meta-analysis or a SWiM to visually report synthesis and directionality of some findings when comparing larger volume to smaller volume of transfusions.

Planned subgroup analysis was not possible given the limited populations in the included studies. As such, we could not conduct subgroup analyses in critically ill, trauma, surgical, cardiac, myocardial infarction or postpartum patients.

Agreements and disagreements with other studies or reviews

This is the first identified systematic review to evaluate two volume strategies of RBC transfusion in a broad and diverse group of hospitalized patients independently of the underlying transfusion threshold used. To date, we have identified only one previous systematic review that evaluated the volume of RBC transfusion on clinical outcomes comparing single-unit versus double-unit red blood cell transfusion in patients with malignant hematological disorders [73]. They included eight studies (two RCTs and six NRSIs) and concluded that in patients with hematological disorders, one-unit RBC transfusion was associated with lower hemoglobin levels at hospital discharge and a reduction in the total number of RBC units used per admission, with no significant difference in terms of length of hospital stay, rebleeding risk, transfusion-associated adverse events and mortality [73]. All full-text studies included in their review were also included in our review, except for one abstract that does not yet have a full-text publication (author was contacted) [74]. Our results are thus in line with this review and align with current practice guidelines.

AUTHORS' CONCLUSIONS

Implications for practice

Analysis of the published evidence to date reveals that transfusing two units of red blood cells confers no added clinical benefit compared to using one unit of red blood cells in hospitalized adults. Despite the very low-certainty evidence for mortality, we suggest adoption of a standardized policy of transfusing a single unit of red blood cells per transfusion event in stable anemic hospitalized adults.

In children and preterm infants, there is insufficient current evidence to support a specific transfusion volume. While red blood cell transfusions in preterm infants have been shown to improve oxygen delivery overall, those with lower hematocrit levels showed signs of physiological benefit, suggesting transfusion thresholds matter [75]. Transfusions in neonates may be under-recognized for complications (transfusion-associated circulatory overload and transfusion-related acute lung injury), so minimizing their use through guidelines, delivery practices, and reduced blood loss is recommended [76]. Pediatric practitioners should consider using smaller transfusion volumes when clinically appropriate, to limit resource utilization, patient exposure to allogenic red blood cell units (we found low-certainty evidence for total red blood cell volume exposure) and potential complications related to volume overload.

Our results align with the ‘Choosing Wisely’ recommendations [9], suggesting that it is likely safe and resource-sparing to reduce the amount of RBC transfused to anemic hospitalized patients.

Equity-related implications for practice

While the studies were conducted in high income countries with higher resources and included a broad range of patients and conditions, it is unclear that these findings fully apply to low- and middle-resource countries. However, sparing costly resources may be especially important for low- and middle-income countries.

Implications for research

It remains unclear if transfusion volumes in hospitalized adults may have an effect on clinical outcomes in specific populations. Further studies may strengthen the certainty of the evidence, but may be difficult to justify, given that larger blood volume exposures do not appear to lead to any difference in clinical outcomes (mortality, length of stay and transfusion-associated adverse events). Further studies are likely warranted in specific populations that may potentially benefit from higher volume transfusions (trauma patients, for example).

In contrast, additional evidence is needed to guide transfusion volume recommendations in children and neonates. In pediatrics, red blood cell transfusion volumes (10 mL/kg to 20 mL/kg per transfusion) are substantially higher than those used in adults (approximately 4 mL/kg per transfusion, based on one unit of red blood cells for a 70 kg person). Consequently, children may face a greater risk of fluid overload, pulmonary edema, or cardiac decompensation, along with the potential overuse of a valuable and limited blood resource, because of current practice. Moreover, transfusion volume requirements in children can vary depending on gestational age, chronological age, and chronic conditions such as cyanotic heart disease or sickle cell anemia. Given the relatively higher risk of ‘overtransfusion’ compared to adults, optimizing red blood cell volumes in pediatric transfusions is an important area of research. Future studies should explore the safety and efficacy of progressively smaller volumes, potentially in the range of 15 mL/kg down to 5 mL/kg, in line with the one-unit policy applied in adult medicine. These studies would also contribute to further equity for children, who are often underrepresented in research.

Equity-related implications for research

More research on red blood cell (RBC) transfusion volume is needed in low- and middle-income countries to determine whether these findings are generalizable to these settings.

SUPPLEMENTARY MATERIALS

Supplementary materials are available with the online version of this article: 10.1002/14651858.CD015898.pub2.

Supplementary material 1 Search strategies

Supplementary material 2 Characteristics of included studies

Supplementary material 3 Characteristics of excluded studies

Supplementary material 4 Characteristics of ongoing studies

Supplementary material 5 Risk of bias

Supplementary material 6 Analyses

Supplementary material 7 Data package

Supplementary material 8 Risk of Bias assessment and judgment for non-randomized studies using ROBINS-I

ADDITIONAL INFORMATION

Acknowledgements

We would like to thank the CHU Sainte-Justine for their ongoing support of our research in transfusion medicine.

We acknowledge the contribution of Laurie Bernier and Li Shu Yin Han for their help with the citation review, as well as Eva Amzallag for her help with the citation and full-text review.

Editorial and peer reviewer contributions

Cochrane Central Editorial Service and Cochrane Hematology supported the authors in the development of this review.

The following people conducted the editorial process for this article:

- Sign-off Editor (final editorial decision): Lise Estcourt, Haematology/Transfusion Medicine, John Radcliffe Hospital, Oxford;
- Managing Editor (selected peer reviewers, provided editorial guidance to authors, edited the article): Sue Marcus, Cochrane Central Editorial Service;
- Editorial Assistant (conducted editorial policy checks, selected peer reviewers, collated peer-reviewer comments and supported editorial team): Cynthia Stafford, Cochrane Central Editorial Service;
- Peer-reviewers (provided comments and recommended an editorial decision): Mark T Friedman, DO, Department of Pathology, NYU Langone Health (clinical/content review); David Mathews, Specialist Biomedical Scientist, Southern Health and Social Care Trust, Northern Ireland (clinical/content review); Brian Duncan (patient and public review); Lindsay Robertson, Cochrane (methods review); Jo Platt, Central Editorial information Specialist (search review).

Contributions of authors

Nadia Roumeliotis wrote the protocol and full manuscript, screened abstracts and full texts, extracted data, conducted risk of bias and GRADE assessment. Nadia Roumeliotis entered the data into RevMan, performed the analyses, synthesized evidence and drafted the manuscript.

George Sabbagh reviewed study protocol study design, screened abstracts and titles identified in the searches, applied inclusion and exclusion criteria, extracted data and reviewed the final manuscript.

Meriem Belkacem screened abstracts and titles, applied inclusion criteria, extracted data, conducted risk of bias, created tables and reviewed the final manuscript.

Philippe Dodin developed the search strategy for the medical literature search and updated review. Philippe Dodin reviewed the final manuscript.

Geneviève DuPont Thibodeau, Jeannie Callum and Marisa Tucci contributed to conceptualization of the study, interpretation of the results and reviewed the full manuscript, and final version, as content and methods experts in the field.

François Carrier reviewed the protocol and full manuscript, reviewed study designs, reviewed risk of bias and conducted GRADE assessment. Francois Carrier interpreted the data and reviewed all analyses.

Jacques Lacroix designed the study, wrote the protocol, reviewed inclusion criteria, outcomes, conducted risk of bias and assessed the study designs of studies identified by the literature review. Jacques Lacroix reviewed the full manuscript and provided insights from practice and research.

Declarations of interest

NR has received grants from the Fonds de Recherche Quebec - Santé (FRQS) and Canadian Institutes for Health Research (CIHR) for research unrelated to this work.

GS, MB, PD, GDT, MT and JL have no declarations of interest.

JC has received a grant/contract from Canadian Blood Services.

FMC reports being the holder of a philanthropic research chair in transfusion medicine (Chaire de recherche en médecine transfusionnelle Fondation-Héma-Québec Bayer de l'Université de Montréal) that promotes best transfusion practices.

JL has no declarations of interest.

Sources of support

Internal sources

- CHU Sainte-Justine, Canada

  Start-up funds for Nadia Roumeliotis

External sources

- None, Other

  No sources of external support received.

Registration and protocol

Protocol (2024) DOI 10.1002/14651858.CD015898 [31]

Data, code and other materials

As part of the published Cochrane review, the following is made available for download for users of the Cochrane Library (see Supplementary material 7: full search strategies for each database (Supplementary material 1). Analyses and data management were conducted within Cochrane's authoring tool, RevMan, using the inbuilt computation methods [43]. Template data extraction forms from Excel are available from the authors on reasonable request.

Disclosure of artificial intelligence use

No artificial intelligence was used in the production of this review, including: screening of citations, data extraction, risk of bias, analyses, manuscript writing, or tables and figures.

History

Protocol first published: Issue 11, 2024

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ADDITIONAL TABLES

Table 1. Table 1. Overview of Included Studies

Study ID Country	Study Design	Setting & site number	Population	Sample Size	Intervention	Comparator	Outcomes of interest
Wong 2005 Canada	RCT, open label 1:1 randomization	NICU – 1 site	Premature infants ≤ 1500 g requiring transfusion	20	20 mL/kg RBC transfusion (N = 10)	15 mL/kg RBC transfusion (N = 10)	Mortality (hospital); RBC units transfused per patient; respiratory adverse events; ROP
Maitland 2019 Malawi/ Uganda	RCT, open-label 1:1 randomization	Ward – 3 sites	Children aged 2 months to 12 years with anemia	3199	15 mL/kg RBC transfusion (30 mL/kg whole blood) (N = 1598)	10 mL/kg RBC transfusion (20 mL/kg whole blood) (N = 1598)	Mortality (48 hours, 28 days, 90 days and 180 days); transfusion-associated adverse event; RBC units transfused per patient; allergic reaction
Hamm 2021 USA	RCT 1:1 randomization	Hospital – Postpartum Unit 1 site	Adult postpartum women with anemia (hemodynamically stable)	66	2 RBC unit transfusion (N = 33)	1 RBC unit transfusion (N = 33)	Composite complication (stroke, TIA, ACS, ICU transfer, or death); RBC units transfused per patient; length of stay; thrombotic event; transfusion reaction
Chantepie 2023 **France**	RCT, open-label 1:1 randomization	Hospital – Hematology Wards - 5 sites	Adults with acute leukemia or undergoing ASCT/ HSCT	245	2 RBC unit transfusion (N = 120)	1 RBC unit transfusion (N = 125)	Composite grade 3 complications including (1 of): stroke, TIA, ACS, heart failure elevated trop, ICU transfer or death; RBC transfusions per cycle; RBC transfusions per patient; hemorrhagic event; transfusion-associated adverse events; fluid overload; thrombotic event; length of stay
Byun 2023 Korea	RCT, randomized equally into 1 of 4 groups, with 2 x 2 factorial	Malignant hematology ward	Adults with newly diagnosed AML, aged 19 to 70 years	77	2 RBC unit transfusion (N = 39)	1 RBC unit transfusion (N = 38)	Length of stay; transfusion-associated adverse event; RBC units transfused per patient; transfusion events
Khodabux 2009 Netherlands	Controlled NRSI 2 units, parallel groups, comparing 2 different volumes of RBC per transfusion	NICU – 2 sites	Premature infants < 32 weeks' gestation with anemia	314	20 mL/kg RBC transfusion	15 mL/kg RBC transfusion	Mortality (hospital); volume RBC units transfused per patient; transfusion events per patient
Berger 2011 Switzerland	Controlled NRSI	Malignant hematology ward – 1 site	Adult (> 16 years) inpatients receiving intensive	139	2 RBC unit transfusion (N = 77)	1 RBC unit transfusion (N = 62)	Mortality (overall); RBC units transfused per chemotherapy cycle; RBC units transfused per event; RBC trans-

Table 1. Table 1. Overview of Included Studies *(Continued)*

	before-after study (2007 versus 2009 after 1-unit RBC practice change)		chemotherapy, autologous/allogenic HSCT				fusion events; severe rebleeding (> grade 3)
Avdic 2016 USA	Controlled NRSI before-after study (2009–2011 versus 2011–2013 after 1-unit RBC practice change)	BMT Unit – 1 site	Adult HSCT patients (autologous and allogeneic)	359	2 RBC unit transfusion (N = 166)	1 RBC unit transfusion (N = 193)	Length of stay; transfusion events per length of stay; RBC units transfused per patient
Chantepie 2017 **France**	Controlled NRSI before-after study (2010–2012 versus 2013–2014 after 1-unit RBC practice change)	Malignant hematology ward – 1 site	Adult (> 16 years) inpatients receiving intensive chemotherapy, autologous/allogenic HSCT	312	2 RBC unit per transfusion (N = 186)	1 RBC unit per transfusion (N = 126)	Mortality (30-day); severe rebleeding (> grade 3); length of stay; RBC units transfused per patient; transfusion events
Lamarche 2019 **Canada**	Controlled NRSI before-after study (2016 versus 2017 after 1- unit RBC order set)	Malignant hematology ward – 1 site	Adults with newly diagnosed leukemia (AML, APL, ALL, MPAL, or HDCT	134	2 RBC unit transfusion (N = 65)	1 RBC unit transfusion (N = 69)	Mortality (hospital and 30-day); length of stay; transfusion-associated adverse event; severe rebleeding (> grade 3); RBC units transfused per patient
Bowman 2019 **USA**	Controlled NRSI before-after study (2013–2013 versus 2014–2015 after 1 unit RBC order set)	Adult Hematology-Oncology Inpatient Unit – 1 site	Adults with hematologic malignancies	273	2 RBC unit transfusion (N = 147)	1 RBC unit transfusion (N = 126)	Mortality (hospital and 30-day); RBC units transfused per patient; transfusion events per admission; length of stay; transfusion-associated adverse events
Marco-Ayala 2024 **Spain**	Controlled NRSI before-after study (2016–2019 versus 2019–2022 after 1 unit RBC practice change)	Hematology wards – 1 site	Adult inpatients receiving ASCT	340	2 RBC unit transfusion (N = 153)	1 RBC unit transfusion (N = 187)	Mortality (30-day); length of stay; RBC units transfused per patient; transfusion events

ACS: acute coronary syndrome, ALL: acute lymphoblastic leukemia, ASCT: autologous stem cell transplant, AML: acute myeloid leukemia, APL: acute promyelocytic leukemia, BMT: bone marrow transplant, HDCT: high-dose chemotherapy, HSCT: hemopoietic stem cell transplant, ICU: intensive care unit, IVH: interventricular hemorrhage, MPAL: mixed phenotype acute leukemia, NICU: neonatal intensive care unit, NRSI: non-randomised studies of interventions, RBC: red blood cell, RCT: randomized controlled trial, ROP: retinopathy of prematurity, TIA: transient ischemic attack.